

WAR MADE NEW

TECHNOLOGY, WARFARE, AND
THE COURSE OF HISTORY,
1500 TO TODAY

MAX BOOT



GOTHAM
BOOKS

INTRODUCTION:

Revolutions in Military Affairs

A sudden tempest which turns everything upside down.

Those plaintive words, uttered by an Italian of the early sixteenth century, could just as easily have come from a Sudanese warrior in 1898 confronting British machine guns, a Frenchman in 1940 experiencing the German *blitzkrieg*, or an Iraqi in 1991 facing American smart bombs. All found themselves in the midst of a military revolution they did not understand—and all paid a heavy price for their backwardness.

The French invasion of Italy in 1494 inaugurated the modern age in which warfare, which had been relatively static for a thousand years, was to change with bewildering and accelerating rapidity. This process would lead some states to domination, others to oblivion. It would profoundly disturb the balance of power first within Europe and then in the rest of the world, giving rise to the Western hegemony that has not been eclipsed even to this day. It would change the very nature of the state itself, providing a powerful impetus for the rise of modern governments and their inexorable expansion into the leviathans of the twentieth century. Its impact is still being felt, as armed forces around the world grapple with the consequences of information technology that is reshaping war in ways that are as profound as they are unpredictable.

From one perspective, it might seem that in warfare, as in many other realms, change is slow and gradual: that it is characterized, in other words, by a process of continual evolution, not by a few wrenching revolutions. This is to some extent true; we must not pretend that change was more sudden or sweeping than it actually was, or unduly emphasize novelty at the expense of continuity, which is always considerable. But in the military sphere, just as in science, economics, art, or culture, change is not evenly distributed across space and time. Sometimes innovations cluster together to produce a major change in the way people live—or, in the case of the military, the way they

die. Obvious examples of transformational technologies include the steam engine in the late eighteenth century and the computer in the late twentieth century, both of which spread from one area to another, transforming everything from production to transportation. When this happens, a *revolution* is said to have occurred.

This book examines four such instances of great change in warfare over the past five hundred years: the Gunpowder Revolution, the Industrial Revolution, the Second Industrial Revolution, and the Information Revolution. Scholars may quibble about exactly how many revolutions have occurred or when they started and ended. But few would deny that at least the first three were periods when new technology combined with new tactics to reshape the face of battle. The verdict on the distinctive nature of the ongoing Information Revolution is not yet in, and some caution is necessary in reaching any preliminary conclusions. But even if the current transformation does not prove as radical in its implications as the preceding revolutionary periods, it has already exerted a substantial impact and one that is likely to grow over time. Indeed, advances in computing, which have made possible precision-guided munitions, stealth aircraft, and stellar surveillance systems, have sparked much of the current interest in what are known as *Revolutions in Military Affairs*.

This term derives from the old Soviet general staff, which, as early as the 1960s, began to speak of a "military technical revolution"—a description first applied to nuclear missiles and then to information technology that multiplied the power of conventional (i.e., nonnuclear) weaponry. The Soviet debate over the impact of these technologies spurred a similar discussion in the United States. By the early 1990s, following the impressive U.S.-led victory in the Gulf War, which served as an advertisement for the power of precision weaponry, it became commonplace to hear heated debate in the hallways of the Pentagon about the "the RMA." (The switch from "military technical revolution" to the "revolution in military affairs" was made to signify that the changes in question involved more than simply new technology.) Radicals suggested that all the old verities of warfare had been outmoded and that sweeping changes were necessary to compete in this brave new world. Reactionaries scoffed that nothing significant had changed and that no revolution had occurred. The debate rages to the present day, not only in the U.S., but also in countries as diverse as Britain, France, China, Russia, and Israel that seek to keep pace with the American armed forces.

Intense as this controversy is, it has been curiously unsatisfying for a historically minded observer. Most books and articles on the current RMA are concerned only with the present and future; most writing on historical RMAs is concerned only with the long-ago past; and studies that take a more comparative approach tend to address only a small audience of specialists.

Moreover, writing on the current RMA is dominated by partisans who seek to either inflate or deflate the importance of information technology, while ignoring the broader political implications of these military transformations.

This book is intended to present a fair-minded account that brings together past and present to offer fresh insights about the future. It does not attempt to construct a definitive model of how military innovations occur; that is a job for social scientists. Nor is it a manifesto calling for a radical overhaul of the world's militaries; that is a job for seers and strategists. This is a work of history that attempts a less ambitious but no less challenging task: trying to tell the story of the past five hundred years of war through the prism of four great revolutions that changed the nature of politics and society as much as they changed combat.

THE ROLE OF TECHNOLOGY

While this book is organized around technical innovations, it is by no means a brief for technological determinism—the theory, held by almost no one in its pure form, that all significant historical developments flow inevitably from the development of certain machines, tools, and weapons. This argument is no more tenable than any attempt to explain all of human history with reference to just one factor, whether that factor is economics, culture, politics, religion, geography, climate, birth order, or the alignment of the stars. Technology itself, it may plausibly be argued, is often not a prime mover but a second-order consequence of food surpluses, urbanization, secularism, political stability, strong patent laws, and other conditions that create a fertile climate for innovation. Moreover, the history of warfare has been profoundly altered by forces, such as the rise of nationalism and democracy, that have little to do with new tools. But just as there is a danger of technological determinism, so there is an equal danger of ignoring the effects of technology and thinking that willing bodies and stout hearts can overcome anything. This was the view of many military strategists prior to World War I. A few years of trench warfare revealed that no amount of *élan* could allow a soldier to outrun a machine-gun bullet or stop an artillery shell. The tools of war do matter.

This book seeks to avoid the Scylla of overvaluing technology and the Charybdis of undervaluing it—the extreme views represented in the opening epigrams from J. F. C. Fuller and Napoleon Bonaparte. I agree with another historian who wrote "that we must steer between two dangerous determinisms: the technological—'what can be done will be done'—and the psychological—'where there's a will there's a way.'" My view is that technology

sets the parameters of the possible; it creates the *potential* for a military revolution. The extent to which various societies and their armies exploit the possibilities inherent in new tools of war and thereby create an *actual* military revolution depends on organization, strategy, tactics, leadership, training, morale, and other human factors. Ultimately, a military revolution, like a scientific revolution, demands a "paradigm shift" from one set of assumptions to another. For this reason, this book will highlight the personalities of important commanders—the likes of Gustavus Adolphus, Helmuth von Moltke, Isoroku Yamamoto, and Curtis LeMay—who struggled to take advantage of new weapons. No technical advance by itself made a revolution; it was how people responded to technology that produced seismic shifts in warfare.

Readers should not be misled by the section and chapter headings which highlight innovations such as "gunpowder," "rifles," and "tanks." The scope of this work is much broader than that. The weapons mentioned are only a shorthand way of referring to more sweeping changes that occurred at the organizational and doctrinal level.

Nor do I mean to emphasize military revolutions to the exclusion of everything else. I will not attempt to challenge most of the theses put forward in a number of prominent recent works that have sought to explain the course of human development. Among the more celebrated are Jared Diamond's *Guns, Germs, and Steel*, which emphasizes the importance of geography, demography, and environment; Paul Kennedy's *The Rise and Fall of the Great Powers*, which stresses economics; and David Landes's *The Wealth and Poverty of Nations* and Victor Davis Hanson's *Carnage and Culture*, both of which focus, in different ways, on cultural factors. In my view, all of these books have a large measure of validity and are not, for the most part, mutually contradictory. Rather than attempting to supplant them, this book will supplement them by highlighting the importance of certain vital military developments in the making of the modern world.

Each section begins with a brief introduction to make clear that what was happening on the battlefield was only part of far-reaching changes occurring in the wider society. Cannons and muskets were only two of many innovations that swept Europe during the Renaissance; machine guns, rifles, and gunboats were only some of the products mass-produced in factories during the Industrial Revolution; the bombers and tanks of World War II were not too dissimilar from civilian airplanes and tractors; and, in our own day, computing power has not only made it possible for the U.S. armed forces to track vast numbers of targets but also made it possible for Wal-Mart to track vast numbers of detergents. Inevitably, there was a lag, ranging from a few decades to a few centuries, between the initial development of a technology and the moment when it transformed the battlefield.

Following each scene-setter, we will examine key battles that show how

the revolution in question played out. In describing these developments, I will seek to avoid the anodyne language typically employed by soldiers and academics. It is all too easy to speak of flanks turned, positions entrenched, battles won and lost, and to lose sight of limbs shattered, heads blown off, bodies reduced to bloody pulp. It is typical to speak of weapons in terms of range and rate of fire, and it is important to do so. But it is also important to investigate the impact of grapeshot on a skirmish line, of a machine gun on a formation of spear-toting warriors, of an incendiary bomb on a neighborhood packed with civilians. Much of the debate over "military revolutions" today has a sterile quality to it, as if the subject under discussion were a better way to manufacture microchips or to deploy a new type of broadband fiber. My intention is to return the focus where it belongs—to the soldier struggling to kill or avoid being killed, and to his commander struggling to master the remorseless logic of carnage.

In the business world, the usefulness of technical innovations can be checked easily by glancing at the stock tables. In the military world, the only test worth anything is the test of battle, and it is here that we must look for the impact of technology upon military affairs.

WHICH BATTLES, AND WHY

There is a long tradition of "battle histories," stretching all the way back to Sir Edward Creasy's 1850 classic, *Fifteen Decisive Battles of the World*. Unlike Sir Edward's selections, or those of J. F. C. Fuller in a later work in a similar vein, the battles described here were not chosen because they were historical turning points, though some were. Nor were they chosen simply because they are interesting and dramatic, though all are. They were chosen with three primary considerations in mind:

- First, to illustrate different aspects of the various revolutions—battles pitting Europeans against one another and against non-Europeans; battles on land, sea, and, in the later stages, in the air.
- Second, to demonstrate the development of the four revolutions over time. Needless to say, technological revolutions, unlike some political ones, do not occur in a day. They may stretch out over decades, even centuries, and change shape as they go. The Gunpowder Revolution started with clumsy arquebuses and wound up with more deadly flintlocks; the First Industrial Revolution began with single-shot rifles and wound up with machine guns; the Second Industrial Revolution began with propeller-driven biplanes and

wound up with jets firing missiles; the Information Revolution began with laser-guided smart bombs and where it will end no one yet knows. The battles described in this book are meant to offer snapshots of these developments at various stages.

■ Finally, to show how one side can gain a decisive edge over an enemy—though not necessarily keep it. What role does technology play, as opposed to training, strategy, leadership, logistics, and other factors?

Readers might wonder why there is no separate heading for the atomic revolution and the concomitant rocketry revolution. Books written soon after 1945 on this subject, the history of warfare, inevitably treated nuclear weapons as a transformative event and speculated feverishly about their impact on the battlefield. The U.S. Army even acted on this advice in the 1950s by creating the Pentomic Army, which fielded atomic artillery shells. As it happens, while the atomic bomb helped to deter some wars from happening (for instance, direct combat between the U.S. and USSR), and helped to limit the reach of some wars that did occur (for instance, the Korean War), its impact on the operational level of war was . . . close to nil. At least so far. Because atomic weapons have not had much direct effect on warfare, they will not receive a separate chapter in this volume, although their implications for the future will be discussed in the concluding pages.

Readers also may be surprised to see many prominent conflicts slighted here. Why isn't there more on the Seven Years' War? The Napoleonic Wars? The American Civil War? The First World War? The Vietnam War? The Six-Day War? It can rightfully be argued that many battles that are not described—whether Lepanto (1571), Rocroi (1643), Yorktown (1781), Trafalgar (1805), Waterloo (1815), Gettysburg (1863), the Marne (1914), Stalingrad (1942–43), or Tet (1968)—had profound political repercussions, often greater than those of the battles that *are* described. I can only reply that this book is not intended to be a comprehensive history of warfare, nor even a comprehensive history of decisive battles. Its focus is strictly on the impact of military revolutions on the modern world. What gets left out are wars and battles that were largely static from the standpoint of tactics and technology, a description that applies to most conflicts through the ages. The focus here is not on grinding, attritional struggles but on technological and tactical breakthroughs that illustrate the transformation of the military art.

Other examples could just as easily have been chosen. I could have chronicled the Battle of the Downs (1639) instead of the Battle of the Spanish Armada; the conquest of northern Nigeria (1902–3) instead of the conquest of the Sudan; the invasion of Poland (1939) instead of the invasion of France. Obviously any selection of a dozen or so battles intended to highlight

military developments over the past five hundred years will leave out a good deal. But to cover everything would be, in effect, to cover nothing.

FOUR REVOLUTIONS

For the *Gunpowder Revolution* (a shorthand way of describing what other historians call “the military revolution of early modern Europe”), I have chosen to look at the Battles of the Spanish Armada (1588), Breitenfeld and Lützen (1631, 1632), and Assaye (1803). The defeat of the Armada heralded the rise of England and the eclipse of Spain, but how were Elizabeth I's commanders able to achieve this feat? By mastery of the emerging technologies of oceangoing sailing ships and heavy cannon. Breitenfeld and Lützen, turning points of the Thirty Years' War, were the greatest victories won by the Swedish King Gustavus Adolphus. His forces, armed with matchlock muskets, pikes, and cannons, introduced many of the organizational techniques that still define barracks life today. Assaye is less known but worth studying for a number of reasons: It was a major early success for Arthur Wellesley, not yet the Duke of Wellington; it was a significant step in the British conquest of India; and, most important for our purposes, it was a demonstration of how European soldiers, armed with flintlock muskets and bayonets and schooled in a “battle culture of forbearance,” could defeat Asian armies that had a major edge in manpower and artillery but a crippling deficit in discipline and tactics.

For the *Industrial Revolution* (which includes the democratization of warfare associated with the French Revolution), I have chosen to look at the Battles of Königgrätz (1866), Omdurman (1898), and Tsushima (1905). At Königgrätz (also known as the Battle of Sadowa), one of the most important stepping-stones toward the creation of the German Reich, the Prussians dealt the Austrian Empire a decisive defeat, primarily because their general staff had figured out how to make effective use of new technologies such as railroads and breech-loading rifles. These technologies, along with others, such as machine guns, quick-firing artillery, and field telephones, were soon incorporated by all the armies of Europe, producing a bloody stalemate on the Western Front between 1914 and 1918.

If the Industrial Revolution did not give one European power a clear advantage over another, at least not for long, it vastly increased the gap between the West and the Rest, making it relatively easy for a handful of Europeans to conquer much of Asia and Africa. To illustrate this aspect of the story, I examine the Battle of Omdurman. It later became famous because of the participation of a young lieutenant named Winston Churchill, but Omdurman

was really notable for General Horatio Herbert Kitchener's industrial techniques, utilizing railroads, gunboats, and machine guns to make the reconquest of the Sudan a foregone conclusion.

One of the few non-Western states able to emulate the industrial warfare of the Europeans in this period was Japan. Its achievement was particularly impressive in the naval realm. In the late nineteenth century, the line-of-battle ship, which had changed little since the days of the Spanish Armada, was transformed by the replacement of sails with steam engines, wooden hulls with iron and steel, muzzle-loading smoothbore cannons with breech-loading rifled guns, and solid shot with explosive shells. Japan, which had only recently ended a long period of self-imposed isolation, took full advantage of these inventions to build a navy that demolished the less capable Russian fleet at the Battle of Tsushima, altering the balance of power in the Pacific for decades.

For the Second Industrial Revolution, which transformed warfare in the 1920s and 1930s and whose repercussions were felt in the 1940s, I will examine the fall of France (1940), the Japanese attack on Pearl Harbor (1941), and the U.S. firebombing of Tokyo (1945). The blitzkrieg is one of the best-known examples of a "military technical revolution"—and one of the most misunderstood by the general public. It is commonly assumed, based on the ease with which German armies overran Poland, Norway, Denmark, the Low Countries, and France, that they possessed a big technological and numerical edge over their adversaries. Nothing could be further from the truth; Hitler actually fielded fewer tanks and aircraft than the British and French, and the quality of the Allied weapons was in many cases higher than the Germans'. The German edge lay in their superior ability to coordinate their forces, and in their high quality of leadership, training, and morale. They figured out how to make the best use of the technology of the day; the Allies did not.

The Japanese, at least initially, also had an edge in morale and training. Moreover, by the time of the Pearl Harbor raid, Japan was a world leader in aircraft carriers, naval aviation, torpedoes, and night fighting. Tokyo fully exploited these advantages to destroy much of the U.S. Pacific Fleet at its anchorage and, during the following six months, to overrun all of the principal Western bases between Hawaii and Australia. Eventually, the German and Japanese advantages were negated, in part by the sheer weight of materiel on the Allied side. But the Allies also possessed some superior weapons, such as the American long-range bombers, which General Curtis LeMay employed with brutal efficiency to bring Japan to its knees even before the two atomic bombs were dropped. All of these weapons systems were enhanced by the use of radio, radar, and other inventions of the Second Industrial Age.

The fourth and final revolution to be examined is the ongoing Information Revolution. To explore its impact, I look at the Gulf War (1991),

Afghanistan (2001), and the Iraq War (2003–5). There is some debate over whether the Gulf War was the last of the industrial wars or the first of the Information Age wars. There is merit in both views. The clash of tank armies in the desert is reminiscent of the great battles of World War II. But the disproportionate impact of "smart" bombs, cruise missiles, satellite navigation systems, and stealth planes suggests that the 1991 conflict rightly belongs to the new era, even if only to its early stages.

The success of precision-guided munitions in Desert Storm convinced many of America's enemies that they could not hope to win a conventional struggle. Al Qaeda tried a different tack with its September 11, 2001, attacks which killed almost three thousand people. This was a spectacularly evil—and spectacularly effective—example of guerrilla and terrorist tactics that have always given conventional armies fits. But the U.S. military showed in Afghanistan that a combination of highly trained commandos and precision weaponry—a high-low mix of technologies—could subdue, at least temporarily, even a land renowned as the graveyard of empires.

The U.S. armed forces produced equally spectacular results in the early stages of the Iraq War, concluding with the fall of Baghdad on April 9, 2003. But the aftermath, with its messy guerrilla war and demanding nation-building, proved a challenge for the high-tech American military. Stealth bombers and nuclear-powered aircraft carriers are not much use for keeping order in Fallujah or Baghdad, leading many to question whether technological supremacy is all that it was cracked up to be. Had a few spectacular victories led America into the kind of strategic overstretch that brought down the German and Japanese empires after their initial successes in World War II? Chapter Twelve will examine the first two years of a conflict that, as of this writing, remains far from over.

The summary of the Information Revolution and Chapter Thirteen move beyond contemporary events and into the realm of the immediate future. I examine where the current military revolution is going and what other revolutions are on the horizon in such fields as robotics, cyberwar, biological and chemical weapons, space systems, and nanotechnology. The Epilogue will survey the lessons of the last five hundred years for clues about what lies ahead.

FIVE THEMES

As this book explores four military revolutions, five major themes will emerge:

First, that technology alone rarely confers an insurmountable military edge; tactics, organization, training, leadership, and other products of an effective bureaucracy are necessary to realize the full potential of new inventions.

For this reason, ever since the rise of modern nation-states in the sixteenth and seventeenth centuries, shifts in military power have been closely associated with shifts in governance.

Second, that countries able to take advantage of these shifts have been history's winners while those that have fallen behind in harnessing military innovations have usually been consigned to irrelevance or oblivion. Thus, each revolution has been accompanied by a shift in the international balance of power.

Third, that even if a country figures out how to harness military power, it still needs the wisdom to know the capabilities and limitations of its war machine and thereby avoid squandering it on impossible projects, as have too many successful innovators from Charles VIII to Adolf Hitler.

Fourth, that even with the best strategy, tactics, and technology in the world, no military revolution has ever conferred an indefinite advantage upon its early innovators. Rivals inevitably copy what they can and come up with tactics or technologies to blunt the effectiveness of what they cannot produce or acquire.

And, fifth, that innovation has been speeding up. It took at least two hundred years for the Gunpowder Revolution to come to fruition (c. 1500–1700); one hundred and fifty years for the First Industrial Revolution (c. 1750–1900); forty years for the Second Industrial Revolution (c. 1900–1940); and just thirty years for the Information Revolution (c. 1970–2000). That means that keeping up with the pace of change is getting harder than ever, and the risks of getting left behind are rising. Today, there is no room for error.

America's early lead in the Information Revolution can easily be lost—it may be being lost already—if it does not stay at the forefront of military developments. Other countries and even subnational entities such as al Qaeda have an opportunity to exert power that would have been unthinkable before the spread of personal computers, cell phones, satellite navigation devices, and other Information Age technologies.

There is no magic formula for coping with the complexities of changing warfare. The course of future developments can be glimpsed only fleetingly and indistinctly through the fog of contemporary confusion. As any reader of H. G. Wells, Jules Verne, or Isaac Asimov must know, few prognostications about the future have ever come true. The best way to figure out the path ahead is to examine how we arrived at this point.

In the case of modern warfare, that means casting our eyes back five hundred years to a time when Europe was just emerging from the Middle Ages and cannons and muskets were as new and unsettling as stealth bombers and smart bombs are today.

The Consequences of the Gunpowder Revolution

Historians differ over how long the Gunpowder Revolution lasted. A good case can be made that the major innovations occurred between 1500 and 1700, though it took time for their effects to spread across Europe and then across the world. To review, the most important developments were:

LAND WARFARE: The mounted knights of the Middle Ages were supplanted by infantrymen armed with missile weapons, first the longbow, then arquebuses and muskets. The spread of artillery made old-fashioned fortresses indefensible but gave rise to formidable redoubts known as the *trace italienne*. Cannons, which had started off as siege weapons in the 1400s, got lighter and more mobile over the centuries, making it possible to wheel them onto the battlefield. As the capabilities of both handheld firearms and cannons steadily improved, pikemen found their numbers reduced. By 1700, when the flintlock musket with a socket bayonet became the standard infantry weapon, they had vanished from the battlefield altogether. The saber remained the primary weapon of the cavalryman following a brief and unsuccessful flirtation with pistols. But cavalry's role was much diminished; horsemen became used primarily for scouting and harrying enemy forces once they were defeated. To take advantage of increasing firepower, Maurice of Nassau and Gustavus Adolphus developed a system of organization, tactics, and drill that harked back to the Roman legions. Disciplined soldiers replaced brave but undisciplined warriors in European armies. In a related development, native-born volunteers and draftees began to displace foreign mercenaries.

NAVAL WARFARE: Galleys powered by oars were supplanted by square-rigged galleons propelled by the winds. The old method of fighting at sea—closing with and boarding the enemy—gave way to artillery duels. Two key

innovations were the development of lidded gunports and four-wheeled truck carriages that made it possible to run cannons in and out of the sides of ships. This made the broadside (all of a ship's guns on one side firing at once) the standard naval tactic. European ships were able to sail around the world as a result of advances in navigation that began with the compass and cross-staff in the 1400s and culminated with the invention in the 1760s of the chronometer to measure longitude. The combination of powerful sailing rigs and heavy artillery allowed European vessels to dominate almost every sea and coastline in the world.

The most important changes of all were common to sea and land warfare. Both realms saw the development of professional, standing armed forces that were larger and more lethal than the ad hoc feudal levies of old. These new armies and navies were directed and funded by centralized state bureaucracies that began to emerge in western Europe in the 1500s.

The rise and fall of nations between 1500 and 1800 is largely a tale of which ones took advantage of these military and naval revolutions and which ones did not. Sheer size or wealth was not a good predictor of military outcomes. In all three of the battles highlighted in this section, the richer and bigger polity lost: Spain (Spanish Armada), the Holy Roman Empire (Breitenfeld and Lützen), and the Maratha Confederacy (Assaye). (There are no reliable economics statistics for Maharashtra, the largest Indian state at the time, but the Indian subcontinent as a whole in 1800 had 19.7 percent of world manufacturing output, to only 4.3 percent for the United Kingdom.) Although it's not unusual to lose a battle and still win a war, in all three cases these defeats were not flukes; they signaled the long-term decline of once-mighty behemoths.

This is not to suggest that being rich or populous doesn't matter. Sustaining power in the long term ultimately requires both soldiers *and* silver, and there are many examples in this period where victory went to the side with—to use the wonderful neologism of Confederate General Nathan Bedford Forrest—the “mostest.” France, for instance, was defeated under both Louis XIV and Napoleon by coalitions of other European states that were no more advanced militarily but could, in aggregate, bring more resources to bear. Sweden, despite its astounding triumphs under Gustavus Adolphus and his successors, faded from the Great Power ranks less than a century later because it was too small to compete against the big boys. The point isn't that population size and economic power are irrelevant. They simply aren't enough by themselves to generate effective military power.

Nor is it enough to manufacture or buy modern weapons. Winning generally requires having modern tactics and a modern organizational structure. This helps explain why Francis Drake and John Hawkins defeated the

Spanish Armada, Gustavus Adolphus defeated the Holy Roman Empire, and Arthur Wellesley defeated the Marathas. In all three instances, the winning side had developed an effective bureaucracy, with appropriate funding, staffing, and leadership, to manage its armed forces. The defeated side, by contrast, lacked an effective command structure and remained wedded to outdated military doctrines—the Spanish to boarding enemy ships, the Imperial armies to the *tercio*, the Marathas to using their cavalry hordes to harass the enemy. All three realized they were behind the times and had to change. The Spanish desperately requisitioned all the cannons they could find before their Armada sailed; the Imperial forces decreased the size of their *tercio* and, at Lützen, adopted the linear tactics of the Swedes; the Marathas formed regular infantry units run by European officers. But none adapted fast enough or completely enough to stave off catastrophic defeat.

Spain and Portugal, along with some non-European states (especially the Ming, Ottoman, and Mogul Empires), did well in the first stage of the Gunpowder Revolution. (Amerindians were the big losers during this period, which coincided with Columbus's discovery of the New World.) But in the seventeenth and eighteenth centuries, northern European nations—Britain, France, the Netherlands, Prussia, Austria, Russia, and, for a time, Sweden—rose to dominance.

A vast literature has been devoted to the subject of why some cultures proved more dynamic than others. Why did technological innovation accelerate in Europe after 1400 while stopping in the Islamic world by 1200, in China by 1450, and in Japan by 1600? And why did technological innovation slow down in southern Europe after 1600 while speeding up in the north? Those questions have been pondered by scholars as diverse and distinguished as Max Weber, Oswald Spengler, Arnold Toynbee, William McNeill, Thomas Sowell, and David Landes, without reaching any definitive conclusions.

For our purposes, the important point is simply that states that were once intellectually curious and technologically innovative—and hence militarily formidable—ossified into bastions of reaction. This was true across many fields, but it was in the military sphere that non-Western societies paid the highest price for their stagnation, because Europeans had now gained the means to project significant amounts of power to the far corners of the globe. The civilizations of East Asia, South Asia, and the Near East were rich enough, populous enough, and sophisticated enough that, unlike the tribes of Africa, Australia, and the Americas, they could have competed with the Europeans on the gunpowder battlefield. Many tried. Few succeeded. Their failure made possible the rise of the West.

The Consequences of the Industrial Revolution

The First Industrial Revolution transformed warfare between the end of the Crimean War (1856) and the start of World War I (1914). To sum up, the principal developments were:

LAND WARFARE: Conscription increased the size of armies. Industrialization made it possible to equip these growing forces. Railroads and steamships made it possible to move them farther and faster than ever before. Telegraphs made it possible to coordinate their movements. General staffs made it possible to plan their operations. And rifled, breech-loading artillery and small arms—including, eventually, repeating rifles and machine guns—made it possible for armies to mete out unprecedented destruction.

The extent of change, sweeping as it was, should not be exaggerated. Once armies moved past the railhead, they were still, in many respects, back in the age of Gustavus Adolphus, moving no faster than men could march and communicating no faster than horses could gallop. And although cavalry greatly declined in importance, because increases in firepower made it suicidal to charge infantry positions, horsemen continued to be used effectively as late as World War I for maneuvering outside the static Western Front.

NAVAL WARFARE: Machine guns played little role at sea because most naval battles took place outside their effective range, but the transformation of artillery had an even bigger impact at sea than on land. The introduction of high-explosive shells fired by rifled guns rendered wooden ships obsolete, while improvements in steam power created a superior source of energy to replace sails. This led to the emergence of steam-driven ships made of steel. The old categories of sailing warships—which were classified according to the number of guns they carried, as first-rates, second-rates, third-rates, and

so forth—were replaced by entirely new classes of vessels: battleships, cruisers, destroyers, gunboats. New underwater weapons—torpedoes and submarines—added another dimension to warfare that previously had been confined to the water's surface.

The electrical telegraph could not be employed to communicate between ships at sea because initially it was dependent on wires. Wireless telegraphs, or radios, introduced at the turn of the twentieth century, were better suited to ships, where their bulkiness and need for large amounts of energy were not nearly as serious a drawback as on land. But early radios were so unreliable, and captains were so afraid of giving away their positions with radio emissions or having their signals jammed, that most resorted to traditional signal flags and semaphores to communicate in the heat of battle.

As noted above, the most immediate and obvious impact of all these developments—along with some others, such as the manufacture of quinine tablets to fight malaria—was to accelerate the European conquest of the world. While a handful of nations and tribes escaped Western occupation during the Industrial Age, the only one that managed to become a Great Power in its own right was Japan. Its rise, crowned at Tsushima, unsettled Western designs for dominance in East Asia and forced the Occidental nations to accept an Oriental state as a peer. The Japanese were under no illusion as to why they were being included in the ranks of the Great Powers. A Japanese diplomat invited to the 1899 Hague Conference commented sardonically, "We showed ourselves at least your equals in scientific butchery, and at once we are admitted to your council tables as civilized men."

Another star was rising on the other side of the Pacific. By the turn of the twentieth century the United States had the biggest and richest economy in the world. For many decades after the Civil War, it had not chosen to translate industrial power into military might. But once it embarked on a naval buildup in the 1890s, the U.S. rapidly constructed a formidable fleet—one that defeated the decrepit Spanish navy in 1898 at the battles of Manila Bay and Santiago Bay as decisively as Japan bested Russia seven years later. The development of the U.S. Army took longer, but it, too, swelled to impressive size and effectiveness in 1917–18 (2.8 million men drafted), before being cut to the bone again in 1919. U.S. influence on world affairs waxed and waned along with the size of its armed forces.

Among European states, Prussia was an early leader in harnessing industrial technology for land warfare and using it to defeat larger rivals—Austria and France. Prussia's successes in the German Wars of Unification led eventually to two world wars whose repercussions are still being felt. The Habsburgs, not the Hohenzollerns, might have dominated the new German state but for their inability to raise effective Industrial Age armed forces of their

own. This failure led to Austria-Hungary's final collapse in 1918, following an even more calamitous military defeat. The Ottoman and Russian empires were other examples of states that had been able to generate impressive amounts of military power in the Gunpowder Age but were unable to modernize their armed forces and society in the late nineteenth and early twentieth centuries, and wound up collapsing as a result. Out of their ruins sprouted a panoply of new states from Poland to the Persian Gulf.

TOTAL WAR

The broader impact of the Industrial Revolution was contradictory: It was both boon and bane. Industrial technology allowed countries to produce more food, more medicine, more clothing, more of everything than ever before, and thus increase their populations without triggering a Malthusian crisis. After an initial, painful period of adjustment, industrialized countries became wealthier than any previous societies in history. But at the same time that industrialization made life better for millions, it also resulted in the deaths of millions. The new technologies could be used to extinguish life as effectively as they could be used to support it.

Industrial weaponry did not cause World War I, except indirectly to the extent that it fostered the rise of Germany. But it did make the conflict, paradoxically, both shorter and far more catastrophic than previous "world wars" such as the Seven Years' War (1756–63) or the French Revolutionary and Napoleonic Wars (1792–1815). The figures boggle the mind: From 1914 to 1918, sixty-three million men were mobilized, eight million were killed, and twenty-two million were seriously wounded or disabled. Millions of civilians also died. Those losses pale by comparison with World War II, which would leave some fifty-five million dead, but they were many orders of magnitude greater than those of any previous conflict. Preindustrial states could not possibly have fed, clothed, equipped, moved—or slaughtered—so many individuals. Germany and France had 20 percent of their populations under arms. Britain mobilized only 13 percent, but this was still far higher than the 7 percent that Napoleon had been able to marshal with the *levée en masse*. Not only were there more soldiers than ever, but each one now had at his disposal more firepower than his ancestors could have imagined. A single machine gunner or artilleryman in 1914 could rain down more death than an entire regiment a hundred years before.

The unleashing of total, industrialized warfare wrought profound and lasting changes across the world. The most obvious consequence of World War I was the toppling of old governments and the creation of fresh grudges

that led, after a twenty-year armistice, to World War II. A more subtle but even more pervasive consequence was to change the nature of government itself.

Laissez-faire structures had survived largely intact for most of the Industrial Age prior to 1914. States had been strengthened and centralized by the military demands of the Gunpowder and Industrial Revolutions, but they remained, for most of the nineteenth century, fairly minimalist affairs. Social welfare programs, such as insurance for accident victims and pensions for the elderly, were pioneered by Germany's Otto von Bismarck in the 1880s, but they were still rudimentary prior to World War I. Even in the most advanced nations, taxes were low (the state spent 13 percent of British GDP in 1910, compared to 37 percent in 2003), and most governmental departments were staffed by a mere handful of clerks.

The demands of total warfare brutally exposed the shortcomings of this approach. All of the major combatants in World War I had greatly underestimated the amount of ammunition that would be expended by modern armies. By early 1915, generals on both sides were complaining about shell shortages, and the old procurement system, with governments contracting with private companies, was found to be inadequate to meet the burgeoning demand. Already governments had nationalized railroads and shipping lines to facilitate mobilization. Now the armaments industry was taken under the state's wing to keep armies fighting.

Before long, other vital commodities, ranging from sugar to cotton to fuel, became scarce due to the Allied naval blockade of the Central Powers and Germany's U-boat campaign against Allied commerce. Governments responded by nationalizing more industries, controlling prices and currencies, and rationing goods. The private sector hardly ceased to exist; in fact, big industrial corporations working in close cooperation with governments made large profits from the war effort. But all across Europe and North America, governments became much more deeply enmeshed in the economy than they had been prior to 1914.

Total government expenditures in Britain, France, and Germany spiked almost 1,200 percent between 1913 and 1918; military expenditures shot up 2,000 percent. Armed forces that, before the war, had consumed no more than 4 percent of national income were now taking more than 25 percent. The expansion of government was not limited to fiscal affairs. Civil liberties were curtailed in all the combatant states. The process was most drastic in Germany, where by 1916 a military dictatorship was in control. But even in more liberal nations, such as the U.S., antiwar activists like Eugene Debs were subject to arrest and summary confinement.

While the Great War spawned centralized control, it also spurred egalitarianism, if not always democracy. The privileges of aristocrats had traditionally

been founded on their domination of the means of war—something that had been especially obvious in the case of Europe's knights and Japan's samurai. Warfare had been somewhat democratized in Europe with the rise of infantry by the 1600s, but the professional armies of the seventeenth and eighteenth centuries remained largely dominated by the nobility. Aristocratic control was broken in France by the French Revolution, but it was reestablished after the fall of Napoleon. While many states adopted peacetime conscription in the nineteenth century, most still kept landed elites in charge—a trend exemplified by Germany, where the world's most modern army was dominated by the Junker nobility.

That hierarchical control was hard to maintain in the face of total war. Old social divisions broke down when millions of men were called to the colors and all classes lived and bled together in the trenches. This ghastly carnage seemed to discredit old institutions such as monarchies and old ways of thinking such as liberalism at the same time that it gave fresh impetus to new ideologies such as communism and fascism. Ancient dynasties—the Romanovs, Habsburgs, Hohenzollerns—fell from power, but even countries that did not see a revolution were transformed. With millions of men called away to the front, the resulting labor shortages gave greater power to the workers who remained. As a result, trade unions saw their membership and power soar during the war. Women also gained more rights as they took jobs vacated by men sent to fight. It was no coincidence that during or shortly after the war the vote was given to women for the first time in most developed countries.

World War I led, then, to the growth of government and the demise of traditional social structures. These trends were to some extent slowed or even reversed after 1918, but no government returned to its pre-1914 level. In the early 1920s, the British and French governments were spending 100 percent more than they had been before 1914. Simply paying off wartime debts meant that tax levels had to stay high. *Laissez-faire* economic policies made a partial comeback in many countries in the 1920s—even in the Soviet Union, where the New Economic Policy introduced market incentives in agriculture—but everywhere after 1929 the Great Depression led to a return to the wartime socialist model. In Britain, the U.S., France, and other liberal democracies, economic crises resulted in the rise of welfare states; in Germany, Japan, Russia and other less liberal countries, to the rise of totalitarian states.

The connection between wartime exigencies and peacetime norms was particularly strong in the dictatorships. Nazism, fascism, and communism all explicitly sought to expand and make permanent the societal mobilization of the 1914–18 period. As one Nazi put it, "The National-Socialist ethos grew entirely logically from the soldier ethics of the World War." The connection between wartime and peacetime was more tenuous in the liberal

democracies but no less real. The National Recovery Administration set up by President Franklin Roosevelt in 1933 was, for instance, modeled on the War Industries Board created by President Woodrow Wilson in 1917.

There was, of course, a world of difference between Stalin's Five-Year Plans and Roosevelt's New Deal, between harsh communism and half-hearted socialism, but both were inspired to some degree by the experience of World War I—a conflict that could never have been waged on such a titanic, transformative scale were it not for the changes in warfare that had occurred in the previous half-century. This was the bittersweet legacy of the Industrial Age.

The Consequences of the Second Industrial Revolution

The three preceding chapters have focused on three of the most important innovations of the Second Industrial Revolution (1917–45)—tanks, aircraft carriers, and heavy bombers—but there were many others. Not everything changed, to be sure, but the application of new technologies, especially the internal combustion engine and the radio, along with new ways of utilizing them, redefined military operations for decades to come.

LAND WARFARE: The most obvious change was the introduction of tanks and other motor vehicles, which helped to restore mobility to a battlefield that had turned increasingly static in the period between the Crimean War and World War I. Armored vehicles were most effective when closely coordinated via radio with ground-attack aircraft, a combination that was at the heart of the Germans' widely copied blitzkrieg. Air-and-armor operations would eventually be used by most of the major belligerents of World War II and also by a number of postwar states. The Israeli Defense Forces proved particularly adept at blitzkrieg-type offensives in 1956 and 1967. The Egyptian and Syrian armies, in turn, had some success with this same approach in 1973 before finally being defeated in the Yom Kippur War. The Indian army staged a lightning offensive of its own in 1971. Its conquest of East Pakistan in just fourteen days gave birth to the new state of Bangladesh.

Airborne operations were another new element of warfare, allowing troops to be moved far behind enemy lines using transport aircraft, gliders, and parachutes. Large-scale airborne operations ranged from the Germans successfully dropping paratroopers on Crete in 1941 to the British and Americans unsuccessfully dropping paratroopers into Holland in 1944 (Operation Market Garden). Victorious or not, airborne troops invariably suffered heavy casualties because they lacked much firepower or mobility once they arrived. For this reason, airdrops generally fell out of favor for

large-scale offensives after 1945; with a few exceptions, such as the U.S. invasion of Panama in 1989, they would be used primarily to infiltrate small groups of commandos or spies behind enemy lines.

Amphibious operations were less novel—Julius Caesar had invaded England from the sea in 55–54 B.C.—but they were put on a sounder basis after the failure of the Gallipoli landings in 1915. In the interwar period, the U.S. Marine Corps studied how to seize advanced island bases. This was mostly a matter of coordinating assaults with supporting fire from sea and air, but it also involved developing new equipment such as shallow-draft landing craft with quick-release ramps. The Marines' 1934 *Tentative Manual for Landing Operations* evolved into the basis of amphibious doctrine not only for the Marine Corps but also for the U.S. Army and Navy. Amphibious warfare proved absolutely vital in the Pacific, where the Marines engaged in a campaign of "island hopping," as well as in the European theater, where Allied troops stormed ashore in North Africa, Sicily, Italy, and finally in Normandy and the south of France. All of the major Allied amphibious attacks succeeded but at heavy cost. (The first thirty minutes of Steven Spielberg's film *Saving Private Ryan* is a harrowing dramatization of the dangers involved in sending unprotected infantrymen against entrenched shore positions.) The ability to insert troops by helicopter or fixed-wing aircraft led to a decline in amphibious operations after World War II. Although U.S. Marines bluffed Saddam Hussein into thinking that they were going to land on the beaches of Kuwait in 1991, their commanders concluded that such an assault would be too costly. The last great landing occurred forty-one years earlier, in September 1950, when General Douglas MacArthur's men disembarked at Inchon to cut off the North Korean advance.

While the means of getting infantrymen into battle had improved during the Second Industrial Age, once engaged they had to make do with weapons that were not terribly novel. The typical grunt carried a rifle roughly similar to the one his father had used in World War I; some were actually issued exactly the same rifles. The Japanese armory was particularly antiquated; their main rifle dated from 1905. The best rifle of World War II was probably the U.S. M-1 Garand, a semiautomatic weapon that could fire eight rounds from a single clip with eight pulls of the trigger. General George S. Patton reportedly called it "the greatest battle implement ever devised." Infantrymen were also armed with handheld submachine guns (such as the American Tommy gun, the British Sten gun, and the German and Russian "burp guns") and crew-served heavy machine guns (such as the German MG34 and MG42). The most innovative weapons issued to infantrymen were shoulder-fired antitank rockets such as the American bazooka and the German *Panzerfaust*.

This was part of a general resurgence of a weapon that had last played a major role in warfare in 1812 when British Congreve rockets had inspired

Francis Scott Key's mention of "the rockets' red glare" in "The Star-Spangled Banner." In addition to the German V-1 and V-2, the Red Army deployed the Katyusha, a short-range, unguided rocket that could be fired from mobile launchers in bursts of more than forty at a time. Artillery also improved with the American introduction of proximity fuses, employing a tiny radar set in the nose cone to set off the shell whenever it got close to a large object. (Previously a shell would go off only if it hit the target—a contact fuse—or at some predetermined point after firing—a time fuse.) Some field guns also became self-propelled, moving forward on armored chassis in an echo of tank warfare.

NAVAL WARFARE: For surface vessels and even submarines there was much continuity between the First and Second World Wars. The battleships, cruisers, destroyers, and submarines of the 1939–45 period were generally bigger, faster, and better armed than their 1914–18 predecessors but not fundamentally different. Indeed, they had not changed much since the Russo-Japanese War of 1905. Yet naval warfare was nevertheless transformed by the introduction of aviation. Fleets that were once built around battleships came to be built around aircraft carriers instead.

Aircraft proved superior not just to conventional surface ships but also, in the Battle of the Atlantic, to submarines as well. German U-boats preying on Allied shipping were foiled through a variety of means including convoying of merchants ships and the use of radar and sonar. But the weapon that proved most effective was an aircraft dropping depth charges. The dispatch of long-range B-24s equipped with the latest radar to patrol the North Atlantic in 1943 helped turn the tide against the U-boats. The proliferation of small escort carriers also allowed air cover for convoys even in the middle of the ocean. Submarines proved more effective in the Pacific, where the vast distances precluded effective patrolling by aircraft and where the Japanese did not develop the types of advanced antisubmarine techniques employed by the Allies in the Atlantic. U.S. submarines took a heavy toll on Japanese merchantmen and warships alike once they managed to fix the problems that bedeviled their Mark 14 torpedo early in the war. "A force comprising less than 2 percent of U.S. Navy personnel," naval historian Ronald Spector would write of U.S. submariners, "had accounted for 55 percent of Japan's losses at sea." The torpedo, whether launched by submarines, surface ships, or airplanes, proved the biggest ship-killer of the war.

AERIAL WARFARE: Aircraft as a component of ground and sea warfare proved indispensable, and victory often went to whichever side was more adept at integrating them into its operations.

Heavy bombers suitable for strategic bombing were utilized mainly by the U.S. Army Air Forces and the Royal Air Force. After briefly flirting with

strategic bombing in the Battle of Britain, the Germans largely abandoned it. The Soviets, Japanese, French, and Italians never embraced it to begin with—a decision that some of them would come to regret. A Japanese naval aviator later wrote, “Had Japan developed such bombers as the B-17, I believe the war would have taken a different course. We did not have a single warplane comparable to these aircraft, and Japan paid a heavy price for this lack.” Germany did develop a four-engine bomber, the Messerschmitt Me-264, dubbed the Amerika because it was intended to bomb New York, but it never got past the prototype stage.

Most of the bombs used in the war were iron canisters filled with high explosives that were guided to their targets by nothing more than wind, gravity, and sheer luck. The standard armaments of all aircraft were machine guns and cannons, supplemented later in the war by rockets. By 1945, jet aircraft began to appear that were much faster than their propeller-driven predecessors, but none was deployed in sufficient numbers to make much of a difference. Nor did the long-range missiles developed by Germany, the V-1 and V-2, affect the war’s outcome. These would be the weapons of the future. Into this category also falls the helicopter, which was still in the prototype stage during World War II but would be employed on a large scale by the French in Algeria in the 1950s and by the Americans in Vietnam in the 1960s.

ELECTRONIC WARFARE: In addition to aerial warfare, the Second Industrial Age added another element to the traditional clash of armies and navies: electronic warfare. Since the dawn of organized warfare, armies had always tried to intercept the plans of the other side, usually by employing spies or capturing messengers. What had historically been a hit-or-miss business was placed on a much more scientific basis with the birth of signals intelligence. The widespread use of radios afforded virtually limitless opportunities for learning the enemy’s secrets, provided their codes could be broken. The British were pioneers at this art. The Admiralty’s Room 40 succeeded during World War I in breaking many of the Imperial German Navy’s communications, but the Royal Navy did not take full advantage of this windfall because its code breakers were not well integrated with operational commands. British code breakers did, however, score a historic success in 1917 when they intercepted and decoded a telegram from German Foreign Minister Arthur Zimmermann offering Mexico the return of the American Southwest in return for making war on the United States. The publication of the Zimmermann telegram helped convince the U.S. to enter World War I.

In World War II, the big British coup was replicating the Enigma machine used to encode most German messages. British scientists also pioneered radar

and sonar. All of these inventions were shared with the United States. Meanwhile, U.S. Navy and Army cryptanalysts on their own had already broken the Japanese naval and diplomatic codes. Ultra and Magic—the code names given to the cracking of German and Japanese ciphers, respectively—gave the Allies an invaluable edge over the Axis, who never developed the same level of success in penetrating Allied communications.

Electronic warfare turned into a game of cat and mouse, pitting teams of scientists and engineers against one another. Many lives depended on the outcome of what became known as the “wizard war.” In 1942 U-boats gained a major edge in their attacks on Allied commerce when the German naval intelligence office, B-Dienst, was able to break the ciphers used by the Americans, British, and Canadians to route transatlantic convoys. This information helped the submarines sink more than 1,600 ships that year. The Allies were operating blind for most of the year because the German navy had changed its Triton cipher. It was no coincidence that the advantage in the Atlantic shifted to the Allies at the end of 1942 when they managed once again to crack the German codes and changed their own so that the U-boats would no longer be privy to information about convoy movements.

Similar back-and-forth battles raged around radar, sonar, and radio-directional navigation equipment. For instance, the RAF was able to bomb Hamburg so effectively in 1943 because it came up with a clever system of dispersing aluminum chaff (known as Window) to disrupt German radar. When the Germans developed alternative methods of detecting incoming bombers (for instance, by homing in on their “Identification, Friend or Foe” radio beacons), they in turn were able to devastate the attacking formations. Ultimately, the Allies won the “battle of the beacons”—and with it the war.

THE BUTCHER’S BILL

Because weaponry reached new heights of destructiveness during the Second Industrial Revolution, it is not surprising that the number of fatalities in World War II—some fifty-five million dead—was greater than in all previous wars combined. That horrific figure is somewhat deceptive, however, for two reasons: first, more civilians were killed than in any other conflict, whether by Allied bombs or Nazi death squads; second, more men fought in World War II than in any other conflict. When looking at *military casualties as a percentage of the entire armed forces*, World War II was considerably less destructive than the average war of the eighteenth or nineteenth century. The United States military, for instance, suffered 6.7 percent dead, wounded,

and sickened in World War II, down from 29 percent for Union forces in the Civil War. (These figures include both battle and nonbattle deaths—i.e., those caused by disease, accident, and suicide.)

The percentage of soldiers killed and injured declined in part because of the increasing tendency to disperse troops to blunt the devastating effects of modern firepower. The average number of soldiers deployed per square kilometer of front declined from 100,000 in ancient times to 3,883 during the American Civil War, 404 in World War I, and just 36 in World War II. (It declined further to 25 in the 1973 Yom Kippur War and has kept on falling since, down to 2 in the 1991 Gulf War.) The high casualties in the wars of the First Industrial Revolution (e.g., the Civil War, the German Wars of Unification, the Russo-Japanese War, and World War I) can be ascribed largely to the fact that weapons' lethality had increased faster than the ability of commanders to adjust their tactics. The increasing use of motor vehicles, airplanes, and radios after 1917 allowed troops to spread out over much greater distances, decreasing their vulnerability to modern firepower.

From a long-term perspective, the lethality of weapons increased roughly two-thousand-fold between the Peloponnesian Wars and World War II, between the days of spears and those of tanks, but the dispersion of forces increased by a factor of over four thousand. The fact that "dispersion has actually increased more rapidly than lethality"—in the words of Trevor Dupuy, who generated these statistics—meant that the impact of modern technology was somewhat blunted. Men remained in control, no matter how destructive machines became.

Advances in medicine also kept fatalities lower than might have been expected, given the rise in weapons' lethality. Battlefield medical care was vastly improved in the mid-nineteenth century by Florence Nightingale and her team of nurses in the Crimean War. Armies thereafter did a much better job of organizing litter-bearers and field hospitals. Following Louis Pasteur's announcement of the germ theory of disease in 1878, surgeons began to sterilize medical instruments and wear gowns, masks, and rubber gloves. Such simple steps helped to stop the transmission of diseases that had once been rampant in operating rooms. The invention of safe and effective anesthesia, in the form of ether or chloroform (first used on a wide scale in the American Civil War), revolutionized surgery by making it possible for surgeons to work more slowly and methodically while lessening the chances that the patient would die of shock. The X-ray machine, invented in 1895, allowed doctors to better diagnose the nature of injuries. The discovery of different blood groups in 1901–2 made possible safer blood transfusions in World War I, which reduced deaths from blood loss. In World War II blood transfusions were employed on a much greater scale by mobile surgical

teams. So, too, were antibiotics like penicillin, discovered in 1928 by Alexander Fleming but not mass-produced until 1943.

The result of all these advances was a rapid falloff in deaths from wounds and especially disease. The U.S. Army in World War II suffered 99 percent fewer deaths from disease than in the Civil War. And the number of soldiers who died of their wounds after receiving medical care declined from an average of 15 percent in nineteenth-century wars to 4.5 percent in World War II. (The figure would fall to 2.5 percent in the Korean War, thanks partly to the introduction of helicopters as aerial ambulances, which allowed casualties to be treated faster than ever before.) In military medicine, as in so many other fields, the German army took the lead, but their work was matched and possibly exceeded by the British and American medical corps.

WHAT PRODUCED VICTORY?

There is a tendency to see the outcome of World War II as foreordained: How could the Axis possibly have prevailed over the much greater industrial resources of the Allies? Richard Overy rightly cautions against this "hind-sight bias" in his invaluable book, *Why the Allies Won*. As he points out, "Economic size as such does not explain the outcome of wars." Indeed this book offers many examples of significant battles—e.g., the Spanish Armada, Assaye, Königgrätz, and Tsushima—won by the side with the smaller economy. World War II further proves the point.

The German armed forces were far more starved of resources in the 1920s than their rivals in Britain or France, yet they were able to develop the blitzkrieg while they could not legally buy a single tank. The Germans *out-thought* their enemies in the interwar period, which is why in 1939–41 the Third Reich was able to *outfight* the countries of Western and Eastern Europe even though they were in aggregate much richer than Germany itself. By 1942 Hitler was in control of most of the economic resources of Europe from the English Channel to the Urals. On paper, at least, this gave the Third Reich the potential to compete against the U.S. and USSR; today the European Union, which encompasses roughly the same area, has a GDP greater than that of the United States. Japan, too, grabbed a vast empire for itself in Asia that should have given it greater ability to hold its own. Yet by 1942 the U.S. was outproducing all of the Axis states combined. The USSR, too, staged a remarkable recovery from its devastating losses in 1941 and was soon outproducing Germany, even though the Germans had overrun two-thirds of its coal and steel industry.

The outcome of the battle of production lines was no more predetermined than the outcome of Midway, El Alamein, or Stalingrad. The Allies produced more tanks, aircraft, and ships because they were more skilled than their enemies at mobilizing their industrial base. Hitler did not even start to put his economy on a full war footing until 1942, and by the time his armaments minister, Albert Speer, was starting to implement his major reforms in 1943, Germany was already losing the war. For most of the conflict the German economy was a mess, characterized by wasteful production of too many models of everything (at one point 425 different kinds of aircraft were being made) and not enough standardization. The Japanese were even more backward. The Soviets were less advanced technologically than the Germans, but they were able to churn out greater quantities of simple and reliable equipment under the aegis of their economic planning agency, Gosplan. The United States was able to combine quality and quantity by using the mass-production techniques pioneered by Henry Ford. American factories under the direction of the War Production Board and other bureaucracies achieved stunning leaps in productivity during the war years. And many of the machines they were producing, ranging from *Essex*-class fast carriers to B-29 heavy bombers, were, by the end of the war, the best in the world.

The growing complexity of warfare in the Second Industrial Age made it imperative to integrate complex systems—not only military but also industrial and scientific—into a smooth-running whole. No country had a perfect track record: The Germans did well with tank warfare and submarines but not with strategic bombing or aircraft carriers; the Americans did well with long-range bombers and aircraft carriers but not tanks. On the whole, however, the Allies pulled off the difficult feat of war management far better than the Axis. Nazi Germany was plagued by the erratic and often irrational decision-making of Adolf Hitler, who fostered an atmosphere of bureaucratic chaos and infighting. While Japan had no single leader of comparable power, it was handicapped by the lack of coordination between its army and navy. The British and Americans, by contrast, set up a Combined Chiefs of Staff Committee that, despite some inevitable friction, capably coordinated their joint war effort. Even Stalin, who often fell prey to the same megalomania and delusions as Hitler, learned as the war went along to defer to his increasingly competent general staff and to such gifted commanders as Marshal Georgi Zhukov.

This underscores a theme running throughout this volume: Having an efficient bureaucracy is the key determinant of whether a country manages to take advantage of a military revolution. Just as England was better organized than Spain in 1588, Sweden better than the Holy Roman Empire in 1631–32, Britain than the Marathas in 1803, Prussia than Austria in 1866, Britain than the Sudan in 1898, and Japan than Russia in 1905—so too the Allies were better organized than the Axis by the end of World War II. The

reason German armies were able to reach the gates of Moscow and Japanese armies the borders of India before being defeated was that the Axis had done a better job of organizing *before* the war. This gave them an important initial advantage that they allowed to slip away through catastrophic miscalculations, which once again goes to show that the early movers in a military revolution are not necessarily the long-term winners.

The German and Japanese failure to make better use of their resources and to set reasonable war aims consigned them to defeat. So total and traumatic was their collapse that after 1945 they were forced to renounce the very idea of power politics and pledge to use force only for self-defense. Britain and France were also losers, even though they were on the winning side. They emerged in 1945 much weakened and unable to hold on to their colonies—a fate they might have avoided or delayed had they been more prepared for war. Britain, Holland, Belgium, and France would have had a good chance of defeating the 1940 Nazi offensive if they had done a better job of building armor and air forces. The British might also have been able to stave off the Japanese onslaught in the Pacific if they had invested more in modern aircraft carriers and naval airplanes. Britain was able to avoid total defeat in 1940–41 partly because of its geography but mainly because it had made some wise investments in the late 1930s—particularly in fighters, radar, and code breaking—that helped to offset some of its deficiencies. Poland, Czechoslovakia, and the other nations of central and eastern Europe were not so lucky. Lacking either a favorable geographical position or modern armed forces, they were enslaved in turn by the Germans and the Russians and did not taste freedom until the 1990s.

The Soviet Union and the United States were the biggest beneficiaries of the Second Industrial Age. Their rise to global power ended the period of Western European dominance that began around 1500. Victory in the Great Patriotic War gave communism enhanced legitimacy inside the Soviet Union and made it a more attractive model for other countries from China to Cuba. Likewise the outcome of the war enhanced the power and prestige of the heavily bureaucratized American government. Federal civilian employment increased nearly fourfold between 1939 and 1945, and even after five years of demobilization the total number of civilians employed by Washington in 1950 was 100 percent higher than it had been in 1939. Total federal expenditures were almost three times higher in 1948 on an inflation-adjusted basis than they had been in 1938. Giant corporations such as Boeing, General Motors, and General Electric, which had grown even bigger to manufacture military materiel, continued to dominate the U.S. economy (indeed, the global economy) after the war.

In other words, World War II reinforced the trends toward statism and corporatism that had been given such a big boost by World War I. The

enhanced power of the U.S. government could be used for ends both good (desegregation) and bad (McCarthyism). In the case of the Soviet government, the ills were much greater (the Gulag) and the benefits much less apparent. What united the two Cold War rivals was that in both the U.S. and USSR, the state reached the pinnacle of its power and popularity in the late 1940s and 1950s, largely based on its success at waging war in the Second Industrial Age. It would take two losing guerrilla struggles—in Vietnam and Afghanistan—to crack the aura of state invincibility.

The Consequences of the Information Revolution

It is tempting to argue, based on the difficulties that U.S. forces have encountered in Iraq (and the similar problems of the Israeli Defense Forces in Lebanon, the West Bank, and Gaza Strip), that the effect of the Information Revolution on warfare has been overstated by some enthusiasts. There is some truth to this—there *has* been an awful lot of hype—but it would be a mistake to go too far in dismissing the results of recent advances. That would be akin to denying the effect of the Industrial Revolution in the early 1900s simply because the British had a lot of trouble subduing the Boers. Improvements in technology have always had a more immediate impact on conventional than on irregular warfare. But that doesn't mean that the art of war has remained static. Indeed the tactics of Islamist guerrillas, who rely so heavily on the Internet, cell phones, and satellite television—all of which barely existed in 1980—show just how much things have changed.

Not all of the changes wrought by the Information Age are obvious at first glance, because the basic military systems of the early twenty-first century look roughly similar to their predecessors of the Second Industrial Age. Military analyst Michael O'Hanlon notes that "basic propulsion systems and designs for aircraft, ships, and internal-combustion vehicles are changing much more gradually than in the early twentieth-century, when two of those three technologies had only recently been invented." The average speed of a U.S. Navy destroyer, for instance, has not increased in the past one hundred years. The U.S. Air Force continues to rely on B-52H bombers last built in 1962. And the Marine Corps still uses helicopters that flew in the Vietnam War. What has been changing with great rapidity since the mid-1970s is the communications, targeting, surveillance, and ordnance technology that can make such "legacy" systems considerably more potent.

LAND WARFARE: Advanced armies are still structured, as they have been since the 1940s, around armored forces complemented by infantry troops who move by armored vehicle, truck, and aircraft. The best tank in the world is probably the American Abrams (of which the U.S. has nine thousand), but the British Challenger II, the German Leopard II, the Israeli Merkava Mk. 4, and the Russian T-80 and T-90 come within striking distance. All modern tanks have stabilized turrets, night-vision capabilities, laser range-finders, and targeting computers that allow them to fight in conditions—on the move or in the dark—that would have stymied earlier models. In addition, composite or reactive armor offers far more protection than in years past, and main guns firing depleted uranium rounds have far more penetrating power. Armored personnel carriers and infantry fighting vehicles, such as the American Bradley Fighting Vehicle and Stryker and the Russian BMP and BTR, are essentially light tanks, some running on wheels, others on tracks, that sacrifice armor and armaments for extra room to carry infantry, command-and-control suites, or other cargo. Self-propelled artillery and rocket systems are also mounted on armored chassis.

Armored vehicles have improved over the years. But so have antiarmor weapons. These range from heavy missiles such as the U.S. Hellfire and Russian Ataka-V fired from vehicles or aircraft to handheld versions such as the U.S. Javelin, the Franco-German Milan, and the Russian Kornet. In addition, even the most advanced tanks can be disabled by other tanks, massive mines, aerial bombs, or artillery shells. The full impact of advances in antiarmor technology has not yet become apparent, because the forces that have fought modern tanks in recent years—Iraqis, Palestinians, Chechens—have not possessed the latest defensive weapons. But the U.S. success in wiping out Iraqi tanks from standoff ranges suggests that, in the constant struggle between offense and defense, the advantage may have shifted against heavy armor.

The U.S. Army is responding to these changes by budgeting at least \$124 billion to develop a Future Combat System that is supposed to replace much of its current armored force with a family of lighter vehicles, manned and unmanned, that will have stealth designs that will make them harder to detect and hybrid-electric engines that will lessen their fuel requirements—one of the chief disadvantages of the gas-guzzling Abrams, because it increases demands on vulnerable supply lines. Future vehicles will feature advanced composite armor designed to deliver more protection than current models for the same amount of weight, but they will rely for protection less on armor and more on locating and destroying the enemy before they are attacked. Critics believe this is placing too much faith in “perfect situational awareness,” and that these vehicles will not be of much use against guerrillas who can strike with no warning.

As usual, the infantryman's tools have changed least of all. A modern soldier has better protection than his forefathers if he wears Kevlar body

armor, but his firepower—which comes primarily from a handheld assault rifle like the M-16 or AK-47 and from a variety of crew-served mortars and machine guns—does not vary significantly from that of a G.I. in World War II. A replacement for the M-16 known as the XM29 is under development but it is hardly revolutionary. In addition to shooting the same 5.56 mm rounds as the M-16 out of one barrel, it will have another barrel that can fire 20 mm high-explosive airburst projectiles to a range of half a mile. These minigrenades will come with embedded microchips that will control when they explode, allowing them to kill enemy fighters who might be lying flat on the ground or hiding behind a berm. Alternatively, nonlethal projectiles like rubber balls could be substituted for crowd-control situations. This is not terribly different from the capability afforded by grenade launchers attached to today's M-16s. Electronic guns that are capable of spitting out a million rounds a minute have also been developed. They might permit a soldier to stop an incoming rocket-propelled grenade with a solid wall of lead. But such weapons are years away from being fielded.

Unfortunately for Western infantrymen, the proliferation of small arms can put low-tech foes on an almost equal footing with the representatives of the most advanced militaries. There are 250 million military and police small arms knocking around the world, and more are being manufactured all the time by at least 1,249 suppliers in ninety countries.

The salvation of Information Age infantry, at least when they are conducting conventional operations, is their ability to use a wireless communications device to call in supporting fire on exact coordinates. It is doubtful that any military force will again enjoy the preponderance of power of a Kitchener at Khartoum, but Americans dropping JDAMs on Afghan tribesmen armed with Kalashnikovs—or even on Iraqi soldiers with outdated T-72 tanks—came close. The American edge decreases considerably, however, when its troops have to deploy for peacekeeping or counterinsurgency operations that leave them exposed to low-tech ambushes.

NAVAL WARFARE: Navies remain divided, as they have been since the dawn of the Second Industrial Age, into aircraft carriers, submarines, and surface ships. The major difference is that blue-water naval competition has disappeared after more than five hundred years. No one even tries to challenge the U.S. Navy anymore on the high seas. Virtually every other navy in the world is little more than a coastal patrol force.

The U.S. has twelve aircraft carriers, nine of them *Nimitz*-class, nuclear-powered supercarriers that can carry more than seventy high-performance aircraft such as the F/A-18E/F Super Hornet. A tenth supercarrier is in the works. No one else has a single one. France has the world's only other nuclear-powered aircraft carrier, the *Charles de Gaulle*, but it's half the size

of the *Nimitz*. Russia has one aircraft carrier, the *Admiral Kuznetsov*, that rarely leaves port, and it has sold another one, the *Admiral Gorshkov*, to India. Britain has three small *Invincible*-class aircraft carriers that are used only for helicopters and vertical-takeoff Harrier jets. France, Italy, Spain, Japan, and South Korea have similar helicopter carriers in the works. These ships are comparable to the U.S. Navy's twelve amphibious assault ships, which transport helicopters, jump jets, and marines.

Whenever they leave port, U.S. capital ships are surrounded by surface and submarine escorts. Twenty-four *Ticonderoga*-class cruisers and forty-five (and counting) *Arleigh Burke*-class destroyers come equipped with Aegis phased-array radar that can track up to nine hundred targets in a three-hundred-mile radius. These surface combatants can also operate on their own or in conjunction with smaller vessels such as frigates and minesweepers.

In World War II, ships that didn't carry aircraft were limited to firing torpedoes or heavy guns with a range of less than thirty miles. Starting in the 1960s some submarines were equipped with intercontinental range ballistic missiles, but their targeting was so imprecise that it made no sense to equip them with conventional warheads. Ballistic-missile subs became a mainstay of nuclear deterrence. The development of accurate cruise missiles starting in the 1970s allowed submarines and surface combatants to hit land targets hundreds of miles away with conventional ordnance. Improvements in torpedo design, including the development of rocket-propelled supercavitating torpedoes, also allow submarines to do more damage in their traditional ship-fighting role.

The U.S. has the world's largest fleet of nuclear-powered attack submarines (54) and nuclear-powered ballistic-missile subs (16). Russia is next, with 37 attack submarines and 14 ballistic missile subs. Britain has 15 nuclear-powered submarines, followed by France with 10, and China with six. Not only are U.S. submarines more numerous but they are also more advanced, the most sophisticated being three 1990s-vintage *Seawolves* described by one defense analyst as "the fastest, quietest, and most heavily armed undersea vessels ever built."

Because of the growing power of each of its vessels and the lack of competitors, the U.S. Navy has consolidated its high seas hegemony even while its fleet has shrunk from almost five hundred ships in the 1980s to fewer than three hundred in the early years of the twenty-first century. The potency of U.S. naval vessels is increased by linking together sensors and weapons systems with a tactical Internet known as FORCEnet.

While the U.S. Navy probably will remain unchallenged in blue waters, it faces greater threats as it gets closer to shore. Here water currents, thermal layers, and various obstacles can interfere with even the most advanced sensors, and a variety of defensive weapons systems lurk in wait.

More than seventy-five thousand antiship missiles are owned by seventy countries. A few are ballistic, but most are of the cruise-missile variety. Their potency was proved in 1987 when French-made Exocets fired by an Iraqi aircraft crippled the USS *Stark*, a Perry-class frigate, killing thirty-seven sailors. Earlier, Argentinian-owned Exocets sank two British ships during the Falkland Islands War (1982). Newer antiship cruise missiles such as the Russian-made *Yakhont*, *Sunburn*, and *Uran* are even deadlier because they have faster speeds, greater stealth capabilities, and more accurate, GPS-enhanced targeting. Russia is selling these missiles to customers abroad and some nations, such as China, are developing their own versions.

U.S. warships have sophisticated defensive systems to guard against air attack: Incoming missiles can be deflected by electronic countermeasures, flares, or chaff, or destroyed by naval aircraft, sea-to-air Standard missiles, or, as a last resort, by rapid-fire, radar-guided Phalanx guns. But, like the *Stark*, a warship could be caught by surprise or overwhelmed by a flurry of missiles coming from different directions.

Even more worrisome from an American viewpoint is the fact that transport ships and fuel tankers which have to replenish a fleet at sea have no protection when they are outside the defensive range of a battle group. They are as vulnerable as supply convoys on the roads of Iraq. Because a supercarrier has only about a three-day stockpile of JP-5 jet fuel (6,500 barrels a day are needed during combat operations), the most powerful warship in history could be rendered useless if its fuel tankers were sunk. The cargo container ships upon which global commerce depends also have no antimissile protection. A country or even a subnational group armed with cruise missiles could wreak havoc on the world's shipping lanes at chokepoints like the Straits of Hormuz or Malacca.

The threat to shipping, civil and military, is increased by diesel submarines. The latest diesel submarines have ultraquiet electric engines that make them hard to detect with sonar, and they are much cheaper to buy or produce than a nuclear-powered submarine. Russia has exported Kilo-class diesel-electric subs to China, India, Iran, and Algeria, among others. China is producing its own *Song*-class diesel submarines in a bid to challenge U.S. naval hegemony by using the same strategy that Germany, with its U-boats, once used to challenge British dominion of the waves. U.S. antisubmarine defenses are quite sophisticated, especially in open waters, but even American sensors can have trouble tracking quiet diesel subs in noisy coastal waters.

Mines, which can be scattered by submarines or other vessels, represent another major threat to shipping. More than three hundred different varieties are available on the world market. They can be triggered by changes in magnetic fields, acoustic levels, seismic pressure, or other factors. Some come equipped with microelectronics that allow them to distinguish between

different types of ships, while others have small motors that allow them to move around. This makes it difficult to certify that a shipping channel is free of mines—it may have been safe an hour ago, but not anymore. Demining technology has lagged behind; the U.S. Navy, for one, has never placed much emphasis on lowly minesweepers. It has paid a price for this neglect. In 1987, during operations to prevent Iran from closing the Persian Gulf, an Iranian mine of World War I design nearly sank the frigate *Samuel Roberts*. Four years later, in the Gulf War, the cruiser USS *Princeton* and the amphibious landing ship *Tripoli* were nearly blasted apart by Iraqi mines. And even a cheap motorboat packed with explosives can pose a significant threat to a modern warship. The USS *Cole*, an *Arleigh Burke*-class destroyer, was badly damaged in such a terrorist attack in the port of Aden, Yemen, in 2000.

All of these threats could be largely negated if U.S. fleets were to stay far out at sea, but they have to approach fairly close to land to launch aircraft or missiles with operational ranges of only a few hundred miles. Moreover, the places where the U.S. Navy is likely to fight in the future are dangerously narrow. The Persian Gulf is only thirty miles wide at its narrowest point, the Taiwan Strait only one hundred miles wide.

To maintain its dominance, the U.S. Navy regularly updates the electronics and weapons aboard its warships even as the hulls and propulsion systems remain unchanged. It also plans to build a variety of unmanned vessels along with a CVN-21 aircraft carrier to replace the *Nimitz*-class, a *Zumwalt*-class DD(X) destroyer to replace *Oliver Hazard Perry*-class frigates and *Spruance*-class destroyers, a CG(X) cruiser to replace the *Ticonderoga*-class cruisers, and a smaller and speedier Littoral Combat Ship with no direct parallel in today's fleet that would focus on clearing mines, hunting submarines, and fighting terrorists in coastal waters. All of these new vessels will have improved defenses and information-processing tools as well as "plug and play" capacity that will allow them to be quickly reconfigured for different missions. They will also incorporate composite materials, stealthier designs, and electric propulsion to make them harder to detect, though an aircraft carrier with a 4.5-acre flight deck can never exactly hide.

Whether all of these warships are truly needed, given the U.S. Navy's already substantial lead over all competitors, remains an open question. A program to develop giant sea bases—perhaps akin to offshore oil platforms—that would allow American ground and air forces to operate overseas might be of greater use, given the growing difficulty the U.S. has had in gaining basing and overflight rights from other countries.

AERIAL WARFARE: Fighters such as the American F-15 and the Russian MiG-29 were designed in the 1970s for air-to-air combat, but this has become almost as rare as ship-to-ship actions. Since the Israelis destroyed

much of the Syrian air force in 1982, and the U.S. and its allies made similarly quick work of the Iraqi air force in 1991, few if any aircraft have been willing to challenge top-of-the-line Western militaries. (The U.S. Air Force has not produced an ace—an airman with at least five aerial kills—since 1972.) That may change with the sale to China of the Russian-built Sukhoi Su-30, whose performance characteristics are said to exceed those of the F-15C, but the F/A-22 Raptor, the F-35 Joint Strike Fighter, and the Eurofighter should restore the Western edge. The odds of future aerial dogfights are, however, slim.

Modern surface-to-air missiles pose a more immediate danger, because they are cheaper and easier to operate. The U.S. and its allies have developed effective methods of neutralizing most existing air defenses. In addition to jammers, radar-seeking missiles, and decoys, the U.S. employs stealth technology, first used on the F-117 Nighthawk, then on the B-2 Spirit, and now on the F/A-22 and F-35. Future aircraft may be designed with "visual stealth" technology to render them almost invisible even in daylight.

No other nation has deployed any stealth aircraft. But advanced sensor networks may be able to detect first-generation stealth planes. The Serbs actually managed to shoot down an F-117 in 1999.

None of the most sophisticated surface-to-air missiles, such as Russia's double-digit SAMs (SA-10, SA-15, SA-20), was available to Iraq, Serbia, Afghanistan, or other states that the U.S. has fought in recent years, but they are being sold to other customers. So are shoulder-fired antiaircraft missiles such as the American FIM-92 Stinger, British Starstreak, French Mistral, Chinese Qianwei-2, and Russian SA-7 Grail, SA-14 Gremlin, SA-16 Gimlet, and SA-18 Grouse. There are at least one hundred thousand such systems in the arsenals of over one hundred states and at least thirteen nonstate groups such as Hezbollah, the Revolutionary Armed Forces of Colombia (FARC), and the Tamil Tigers. The best models have a range of 23,000 feet.

The potential of hand-carried missiles was demonstrated in the 1980s when Stingers took a significant toll on Soviet aircraft in Afghanistan. The threat is serious enough for the U.S. to rely increasingly on unmanned drones for high-risk missions and to mandate that manned aircraft in war zones stay above 15,000 or 20,000 feet. SAMs pose an especially great threat to helicopters, which don't have the option of flying that high, and for airplanes taking off or landing. Three cargo aircraft leaving Baghdad International Airport have been seriously damaged by missiles, and, while all of them survived, several U.S. helicopters hit with SAMs in Iraq and Afghanistan did not. An Israeli jetliner was almost shot down in Mombasa, Kenya, in 2002 by al Qaeda operatives firing an SA-7. Only the terrorists' targeting error prevented the deaths of 271 passengers and crew. Other civilian airliners are sure to be less lucky.

Assuming that warplanes can reach their destination, the growing precision of bombs and missiles has made it possible to take out targets with fewer and smaller munitions than ever before. (The U.S. Air Force's latest bomb carries only fifty pounds of explosives.) Weapons are getting smarter all the time. The U.S. Sensor-Fuzed Weapon, first employed in the Iraq War, disperses forty "skeet" antiarmor warheads that use infrared and laser sensors to find and destroy armored vehicles within a thirty-acre area. The Tactical Tomahawk, which entered production in 2004, can loiter up to three hours while searching for targets and receive in-flight retargeting instructions.

The U.S. preponderance in smart bombs and missiles helps to compensate for the relatively small size of its manned bomber force. As of 2005, the U.S. Air Force had only 157 long-range bombers (B-52s, B-1s, B-2s), a considerable reduction not only from World War II (when the U.S. had 34,780) but also from the end of the Cold War (360). While few in number, each B-2 can perform the work of thousands of B-29s by "servicing" eighty "aim points" per sortie.

Tankers such as the KC-10 and KC-135 vastly extend the range and effectiveness of combat aircraft. Cargo-lifters like the U.S. C-5, C-17, and C-130 and the Russian An-70 and An-225 also perform a valuable, if unglamorous, role in projecting military power around the world. The U.S. owns 740 tanker aircraft and 1,200 cargo aircraft, far more than any other country. A lack of such support aircraft makes it difficult for even the relatively sophisticated European militaries to move their forces very far.

A host of other aircraft, ranging from JSTARS and AWACs to Rivet Joint and Global Hawk, perform surveillance and electronic-warfare missions in support of combat forces. Their numbers have been growing: While there were only two JSTARS in the Gulf War, in the Iraq War there were fifteen. But commanders have become so dependent on these systems that there never seem to be enough to go around. These, too, are vital U.S. assets that few other nations possess.

SPACE WARFARE: A growing amount of surveillance, communications, and intelligence work is being performed by unmanned aircraft and satellites. In 2001 the U.S. had an estimated 100 military satellites and 150 commercial satellites in orbit, as much as the rest of the world combined. The U.S. spends more than \$15 billion a year on space, some 90 percent of the global total. The most advanced U.S. surveillance satellites are said to be able to pick out a six-inch object from 150 miles above. A new generation of satellites utilizes stealth technology to make it more difficult for other countries to track their progress and thus to hide equipment from American eyes.

The advantage the U.S. military derives from mastery of space is slowly eroding, however. The Global Positioning System developed by the Defense Department is now widely available for countless commercial applications that have spawned a \$30-billion-per-year industry. A potential enemy could use GPS signals to locate targets in the U.S. the same way the U.S. military uses it to locate targets in Iraq or Afghanistan. The U.S. could jam or degrade GPS signals in wartime, but it would have to do so very selectively for fear of imposing a severe toll on the economy, because GPS devices are now essential for civil aviation, shipping, and other functions. In any case, the European Union in cooperation with China is launching its own GPS constellation, known as Galileo, that would be outside of U.S. control.

This is part of a trend of more and more countries—at least forty to date—lofting their own satellites. But getting access to space no longer requires having your own satellite. A growing number of private firms with such names as Google Earth, Keyhole, DigitalGlobe, and Space Imaging sell or give away high-resolution satellite photos via the Internet. The best of these offer imagery of sufficient quality to identify objects eighteen inches wide. The Israeli-owned ImageSat International offers customers the opportunity to redirect its EROS-A imaging satellite (launched in 2000 aboard a Russian rocket) and download its data in total secrecy with few if any restrictions. Its CEO boasts, "Our customers, in effect, acquire their own reconnaissance satellite . . . at a fraction of the cost that it would take to build their own." The private satellite industry is becoming so pervasive that the U.S. military has turned to it to provide some of its own imaging (typically low-resolution pictures used for mapping) and much of its communications needs.

Targets identified from space could be attacked either with terrorist (or commando) strikes or with the growing number of missiles proliferating around the world. More than two dozen nations have ballistic missiles and by 2015 at least a dozen will have land-attack cruise missiles. Either type of projectile could be topped with chemical, biological, or nuclear warheads. At least eight countries already have nuclear weapons and more are trying to get them, in part to offset the tremendous U.S. advantage in conventional weaponry.

In response, the U.S. is working on a variety of missile defenses. The most advanced are the ground-based Patriot Advanced Capability 3 and the sea-based Standard Missile 3, which already have been deployed to protect U.S. troops overseas. The deployment of a long-heralded system designed to protect the U.S. homeland against long-range missiles began in 2004 with the installation of interceptors in Alaska. Eventually, the U.S. plans to field a multilayered defense using a variety of sensors and weapons on land, sea, air, and space. Also in the works are systems designed to defeat low-flying cruise missiles, which are hard to distinguish from ground clutter.

FOG FORECAST

The changes wrought by the Information Revolution are still in their early stages and have serious limitations. Even the best surveillance systems can be thwarted by simple countermeasures like camouflage, smoke, and decoys, by bad weather, or by difficult terrain like the deep sea, mountains, or jungles. Sensors have limited ability to penetrate solid objects, so they cannot tell what is happening in underground bunkers such as those that North Korea and Iran use to hide their nuclear weapons programs. Urban areas present a particularly difficult challenge: There are far more things to track (individuals) and far more obstructions (buildings, vehicles, trees, signs) than at sea or in the sky—and figuring out whether a person is a civilian or an insurgent is a lot harder than figuring out whether an unidentified aircraft is a civilian airliner or an enemy fighter. It is harder still to figure out how hard enemy soldiers will resist or what stratagems they will employ. No machine has yet been invented that can penetrate human thought processes. Even with the best equipment in the world, U.S. forces frequently have been surprised by their adversaries.

Some strategists expect that advances in information technology will greatly diminish if not altogether obliterate some of these difficulties. The Pentagon is creating a Global Information Grid that will pool data from all U.S. assets, whether an infantryman on the ground or a satellite in space. The ultimate goal: to provide a perfect operational picture—a “God’s-eye view” of the “battlespace.”

This ambitious objective could be furthered by the development of better microwave radars that could see through walls, foliage, or soil; cheaper, more pervasive sensors that could provide 24/7 coverage of the battlefield; better data compression and transmission techniques that could allow more bytes to be sent much faster; and more powerful computers that might make it possible to create, for example, a real-time, three-dimensional model of a city showing all the people who reside in it. Not only are electronic computers getting more powerful in accordance with Moore’s Law, but even greater leaps in computing power may be achieved using biological computers that rely not on electronic components but on DNA, individual molecules, or proteins; optical computers that use light rather than electricity; and quantum computers that exploit the atomic properties of matter. Some scientists speculate that by using such methods a billionfold increase in computing power is possible. “That means you could hold the power of all earth’s present computers in the palm of your hand,” says Stan Williams of Hewlett-Packard.

Yet no matter how far information technology advances, it is doubtful that the Pentagon will ever succeed, as some utopians dream, in “lifting the

fog of war.” The fallibility of American soldiers and the cunning of their enemies will surely continue to frustrate their best-laid plans. Indeed, America’s growing reliance on high-tech systems creates vulnerabilities of its own: Future enemies have strong incentives to attack U.S. computer and communication nodes. Strikes on military information networks could blind or paralyze the armed forces, while strikes on civilian infrastructure, such as banking or air-control systems, could cause chaos on the home front.

Safe to say, adversaries will figure out ways to blunt the U.S. informational advantage. From Operation Anaconda in Afghanistan to numerous misadventures in Iraq, they already have.

AMERICA’S UNPARALLELED POWER

While various setbacks in the war on terror underscore the limits of American power, it is important not to lose sight of the bigger picture: The dawn of the Information Age has coincided with a period of American global hegemony. Part of the explanation lies in U.S. economic strength and its “soft power.” But Europe and Japan are rich, too, yet their global sway lags far behind. What they lack is America’s military strength.

In the early years of the twenty-first century the United States enjoys a preponderance of military power greater than that of any other nation in history. Rome was rivaled in land power by Carthage, Persia, and other states, and its sphere of control never extended far beyond western Europe and the Mediterranean region. Britain was rivaled in naval power by the Netherlands, France, and later Germany, and it was never dominant on land. Today America is rivaled in land, sea, and air power by . . . no one. Although the dominance of U.S. forces can still be challenged when they come into close contact with the enemy on his home turf, they are undisputed masters of the “commons” (sea, air, space), which allows them to project power anywhere in the world at short notice. In the words of journalist Gregg Easterbrook, “The American military is now the strongest the world has ever known, both in absolute terms and relative to other nations; stronger than the Wehrmacht in 1940, stronger than the legions at the height of Roman power.”

What accounts for this huge military edge? Not manpower. The U.S. armed forces in 2006, with about 1.4 million active-duty personnel and 1.2 million reservists, were hardly the biggest in the world. NATO’s European members and China each had roughly 2.2 million men and women under arms. Other nations—North Korea with 1 million active-duty soldiers and 4.7 million reservists, Russia with 960,000 on active duty and 2.4 million in the reserves—were not far behind. In any case, because the U.S. armed

forces typically do their fighting overseas, they seldom enjoy a numerical advantage in their theater of operations. Whether in Iraq or Afghanistan, they have usually been outnumbered.

In the Information Age the U.S. edge lies in quality, not (as it did during the Second Industrial Age) in quantity. Both its soldiers and their equipment are among the best on earth. The U.S. armed forces are not the only Information Age military—Britain, France, Japan, Singapore, Australia, Sweden, and other nations have followed in its wake—but they have gone the furthest fastest.

Among other nations, Israel's success in harnessing the Information Revolution has been particularly consequential. The balance of power in the Middle East shifted sharply in Israel's favor following the Yom Kippur War, because it modernized its military to a far greater extent than its neighbors did. This helps to explain why no Arab nation has mounted a conventional attack on Israel since 1973.

The United States' success in harnessing the Information Revolution produced its profoundest effects without a shot being fired. Aware that the Soviet Union could not keep up with U.S. military advances in the 1980s, Mikhail Gorbachev responded with perestroika and glasnost, reform efforts that spun out of control and ultimately consigned the Soviet Union to the ash heap of history. The demise of the USSR set the stage for the U.S. military dominance that has underpinned the current era of globalization, which has been remarkably devoid of conflict between states.

Wars between states have declined in size and frequency for a variety of reasons, not the least being the integration of old adversaries like France and Germany into the European Union, the international norms propagated by the United Nations Charter, the spread of capitalism and democracy across the world, and the deterrence provided by nuclear weapons. But surely another part of the explanation is that few states feel free to challenge the Pax Americana. Several of those that have tried, including Panama, Serbia, Afghanistan, and Iraq, have seen the downfall of their regimes. "War, let alone 'decisive war' between major states, currently is enjoying an off season for one main reason," writes British strategist Colin Gray. "So extreme is the imbalance of military power in favor of the United States that potential rivals rule out policies that might lead to hostilities with the superpower."

ACCESS DENIED?

It is doubtful that any country will mount a full-spectrum challenge to U.S. military capabilities in the foreseeable future. The entry barriers are simply too high, especially for air, sea, and space systems. *Virginia*-class nuclear

submarines cost \$2.4 billion, *Nimitz*-class aircraft carriers go for \$6 billion, and the F-35 Joint Strike Fighter program will cost at least \$245 billion. The U.S. spends around \$500 billion a year on its military, almost as much as the rest of the world combined. In fact, the U.S. spends more simply on the research, development, testing, and evaluation of new weapons—\$71 billion in 2006—than any other country spends on its entire armed forces. (By way of comparison, the top three spenders after the U.S. are Russia, whose defense budget in 2003 was estimated at \$65 billion; China, at \$56 billion; France, at \$45 billion; and Japan and the United Kingdom, at \$42 billion each.)

It is not only U.S. hardware that's hard to replicate; so is the all-volunteer force that makes it work. Operating high-tech military equipment requires long-service professionals, not short-term conscripts. Countries as diverse as Vietnam, China, Germany, and Russia are emulating the Anglo-American model by downsizing their forces and relying less on draftees. Many other nations have abolished the draft altogether. The U.S. military's edge lies not simply in recruiting high-quality personnel but in its methods for training and organizing them. Initiatives undertaken in earlier decades, such as setting up realistic training centers to simulate combat conditions and forcing the services to work more closely together (the Goldwater-Nichols Act), continue to bear fruit. Few other armed forces have made comparable reforms. Writes one scholar: "The superb U.S. all-volunteer military force . . . is a unique human and institutional asset that less capable foreign rivals can neither copy nor steal."

But a potential adversary does not need to duplicate U.S. force structure in order to challenge it. The United States faces a growing "asymmetric" threat both from other states and from substate groups.

As we have seen, a variety of off-the-shelf missiles can threaten U.S. tanks, surface ships, and aircraft, especially when they get close to hostile territory. The power of smart munitions is outstripping the protection afforded by speed or armor. After 2010, write defense analysts Michael Vickers and Robert Martinage, "the survivability of aircraft carriers, high-structure surface combatants [e.g., tanks], and non-stealthy aircraft of all types could increasingly be called into question as maritime, over-the-horizon 'area denial' capabilities and extended-range air defense systems continue to mature."

Also vulnerable are the ports, airfields, and bases that the U.S. uses to project its power overseas. Imagine how much damage Saddam Hussein could have done in 2003 if he had been able to annihilate the one port in Kuwait that was being used to disembark coalition troops or the large desert bases in Kuwait where over one hundred thousand British and American troops gathered prior to the invasion of Iraq. The Pentagon's 2001 Quadrennial Defense Review warned that "[f]uture adversaries could have the means

to render ineffective much of our current ability to project military power overseas."

If the U.S. armed forces could not count on safe, assured access to overseas bases they would have to radically change the way they do business. It would no longer be practical to rely on large land armies or lots of short-range combat aircraft operating out of vulnerable forward bases supplied by equally vulnerable cargo ships, trucks, and aircraft. The U.S. Army might be forced to rely on a small numbers of commandos supported by long-range aircraft and missiles—the Afghanistan model. The navy might have to depend more on submarines and the air force on stealth aircraft. All the services might have to make greater use of unmanned vehicles. The battlefield, which has been growing less crowded for centuries, might empty out even further as small units try to conceal themselves from ubiquitous sensor networks, emerging only briefly to launch lightning strikes before they go back into hiding.

This has become known as the "swarming" scenario, and it has attracted support from the likes of military historian Alexander Bevin, who writes, "Large concentrations of troops and weapons are targets for destruction, not marks of power, and [in the future] they no longer will exist. . . . Military units, to survive, must not only be small, but highly mobile, self-contained, and autonomous."

Even if these predictions are accurate, however, no timeline is attached. Timing matters. After all, everyone knows that a bull market will be followed by a bear market, but the key to making money in stocks is knowing when that transition will occur. Likewise, the key to winning future wars is knowing when to move from one form of military to another. A premature decision to change (such as the U.S. Army's Pentomic design in the 1950s) can leave one unprepared to fight and win wars that actually occur, Vietnam being the classic example.

In any case, it is doubtful that a complete shift to "swarming" will ever take place. Winning wars, as distinct from winning battles, will continue to require controlling territory, which in turn will require a substantial presence of ground troops, as the U.S. has learned in Afghanistan and Iraq. No wonder weapon will alter this fundamental reality.

"AMERICAN HIROSHIMA"

Even as strategists look to the future, armed forces must not lose sight of the threats of the moment, and they do not come for the most part from traditional militaries. They come largely from terrorist groups—some with state

sponsorship, others without—that utilize the fruits of modern technology to their advantage.

"Irregular" attacks carried out by tribes, clans, or other nonstate actors are as old as warfare itself; they long predate the development of modern armed forces and the nation-state. The religious fanaticism that animates so many of today's terrorists and guerrillas is equally ancient. But technological advances have made such attacks far more potent than in the past. The progeny of the Second Industrial Revolution—assault rifles, machine guns, mortars, rocket launchers, land mines, explosives—long ago spread to the remotest corners of the globe. Fighters who a century ago might have made do with swords and muskets now have access to cheap and reliable weapons, such as the AK-47, capable of spewing out 100 bullets a minute. More advanced technologies, from handheld missiles to chemical, biological, and nuclear weapons, give even a small group of insurgents the ability to mete out far more destruction than entire armies could a century ago. And thanks to modern transportation and communications infrastructure—such as jumbo jets, the Internet, and cell phones—insurgents have the capability to carry out their attacks virtually anywhere in the world.

September 11, 2001, showed the terrifying possibilities of such unconventional warfare. It is easy to imagine that in the future super-terrorists will be able to kill hundreds of thousands, even millions, with effective weapons of mass destruction. All of the materials, as well as the know-how, needed to craft such devices are all too readily available.

Chemical and biological threats will be discussed in Chapter Thirteen, but nuclear weapons, more than any other kind, have the ability to trump U.S. military hegemony. The atomic bomb is more than sixty years old. It belongs to an age of rotary-dial telephones and fin-winged cars. It is a miracle that it has not been used in anger since 1945, but that streak won't last forever. And while Information Age technology offers a reasonable chance of stopping a nuclear-tipped missile, there is much less probability of stopping a terrorist with a nuclear suitcase. (Yes, nuclear bombs have gotten smaller, just like computers.) There is little in theory to prevent al Qaeda from carrying out its oft-expressed desire to create an "American Hiroshima." In the words of a retired four-star general who once ran antinuclear terror programs for the Department of Energy: "It is not a matter of if; it's a matter of *when*."

The most important challenge for the armed forces of the U.S. and its allies in the post-9/11 world is to "leverage" their advantage in conventional weaponry to deal with an unconventional threat. Information technology can be an important part of this task. Embedded microchips can track the 18 million cargo containers moving around the world and prevent terrorists from using them to smuggle weapons. Computerized cameras scanning a crowd may be able to pick out a terrorist based on facial recognition patterns.

Doglike sniffing machines may be able to recognize suspects by their body odor. Powerful computers utilizing artificial intelligence programs can sift vast reams of digital data to pick out information about terrorist plots—if concerns about violating the privacy of innocent people do not get in the way. A variety of unobtrusive sensors can warn of the presence of explosives or chemical, biological, or nuclear weapons. Handheld computer translating devices such as the Phraselator already in use by U.S. troops can bridge some of the language gap between Western operatives and the regions where they operate.

But in the final analysis, having the best technology is not enough to defeat the most ruthless terrorists. In fact, most of the expensive weapons systems being purchased by the U.S. and its allies are almost completely irrelevant to the war against terrorism. Smart bombs can be useful for killing the enemy once he's been located. But figuring out who the enemy is, where he is, and what he is up to—that requires smart *people*.

Various remedies have been proposed to fix the obvious American weakness in this area: Requiring more foreign-language training for the armed services. Recruiting more spies. Setting up a military school devoted to civil governance. Creating an agency tasked with "nation-building." Appointing "cultural scouts"—an enhancement of the existing Foreign Area Officer program, which is widely seen as a career dead end—who would spend long periods of time in foreign lands. These are all good ideas and some are beginning to be implemented, but they are harder to enact than a decision to buy a new weapons system. They require changing the culture of the Defense Department and other government agencies, which have not stressed skills in counterinsurgency, human intelligence, or nation-building.

TOOTH AND TAIL

While the U.S. armed forces have made some major strides in recent decades, especially in getting different branches to work more closely together, they still have a long way to go before they have an organizational structure that makes the most effective use of their high-tech equipment. The army, although putting greater emphasis on more nimble, four-thousand-person brigades, still maintains a divisional structure that dates back to the First Industrial Age, while the basic units of the navy (carrier groups) and the air force (squadrons and wings) date back to the Second Industrial Age. All of the services employ a personnel system developed more than one hundred years ago that treats individuals as interchangeable cogs in a giant machine. Army officers have traditionally spent an average of only eighteen months at

each duty station during the course of a twenty-five-year career, making it difficult to build true expertise in such complex areas as foreign cultures.

Another problem is a cumbersome procurement process that costs too much and takes too long to deliver new equipment. The U.S. paid a heavy human cost for those shortcomings in Iraq, where body armor, armored Humvees, and IED-jamming devices were slow to arrive. The time and expense involved in producing new weapons systems has increased severalfold since the 1970s, even as the private sector has been able to produce products like cell phones and microchips faster and more cheaply. "In order to stay ahead of adversaries with access to technologies available in the global marketplace, the DOD [Department of Defense] needs to shorten the time-frame from concept to testing," writes Lieutenant Colonel Augustus Way Fountain III, a chemistry professor at West Point. That's exactly what the best-run companies—the Microsofts and the Wal-Marts—have done, but the U.S. armed forces (and most others, for that matter) have lagged behind.

In general, the U.S. military is hindered by a sluggish, bloated bureaucracy that has resisted countless reform efforts. "We're No. 1 in the world in military capabilities," says David M. Walker, head of Congress's Government Accountability Office. "But on the business side, the Defense Department gets a D—giving them the benefit of the doubt. If they were a business, they wouldn't be in business."

Some statistics buttress the point. The "tooth to tail" ratio in the U.S. military—the number of support personnel as opposed to actual trigger-pullers—varies from 12:1 for the Army and Marine Corps, to 15:1 for the Navy, and 32:1 for the Air Force. "An astounding 70 percent of the defense budget is spent on overhead and infrastructure (the bureaucratic 'tail')." Only 30 percent directly reaches our combat forces in the field (the 'tooth')," writes retired Admiral Bill Owens. "No community would tolerate 7 out of every 10 police officers sitting at their desks pushing paper. The nation should not tolerate such a ratio in the military."

What makes this unwieldy setup so intolerable is that it can stymie the potential of even the most advanced information technologies. Retired Marine Colonel T. X. Hammes notes that

most commanders must submit their intelligence requirements up the chain of command. Each level validates, consolidates, and prioritizes the requests, which are then fed through the centralized staff system to task the assets that will actually collect against the requests. The information is collected, passed to another section for analysis, then put in the form of a usable product, and finally disseminated through the same cumbersome system. Thus, the premier benefit of the Information Age—immediate access to current intelligence—is nullified by the way we route it through our vertical bureaucracy.

The good news, from the American perspective, is that many potential enemies (e.g., China or North Korea) are encumbered by even more sclerotic bureaucracies. The bad news is that this is not true of nimble, networked groups like al Qaeda. To fight them effectively, the U.S. military will have to display more of the decentralized decision-making that it showed in Afghanistan in the fall of 2001. This will not be easy to do because modern command and control technology is a two-edged sword: It can be used to centralize or decentralize. It will always be difficult for a senior military or political leader to resist the temptation to micromanage operations from afar—a style of leadership that modern communications technology has made easier, but no more effective, than in the past.

Veteran journalist Robert Kaplan rightly calls “the dinosauric, vertical bureaucracy of the Industrial Age . . . the greatest single impediment to America’s ability to wage a successful worldwide counter-insurgency.” Unless the U.S. government can streamline its Industrial Age bureaucracy and become a networked organization, it may find that even purchasing the latest and best technology will not offer sufficient protection against the country’s foes.

EPILOGUE:

Five Hundred Years and Counting: What the Past Teaches About the Future

Any one of the innovations described in the preceding chapter has the potential to radically change the nature of warfare and with it the nature of the international system. While the U.S. has been dominant so far in the Information Age, there is no guarantee that its streak will continue. A challenger, whether a rival state like China or even a nonstate group like al Qaeda, could utilize new ways of war (or, in the case of nuclear weapons, not so new) to alter the balance of power. Cheap to produce and easy to disseminate, germs, chemicals, and cyber-viruses are particularly well-suited for the weak to use against the strong. If any of them become common and effective tools of warfare, especially terrorist warfare, the U.S. and its allies could be in deep trouble.

History is full of examples of superpowers failing to take advantage of important Revolutions in Military Affairs: The Mongols missed the Gunpowder Revolution; the Chinese, Turks, and Indians missed the Industrial Revolution; the French and British missed major parts of the Second Industrial Revolution; the Soviets missed the Information Revolution. The warning that appears at the bottom of mutual fund advertisements applies to geopolitics: *Past performance is no guarantee of future returns*. The end can come with shocking suddenness even after a long streak of good fortune.

Perhaps *especially* after a long streak of good fortune. The longer you are on top, the more natural it seems, and the less thinkable it is that anyone will displace you. Complacency can seep in, especially if, like the United States, you enjoy power without peer or precedent.

Israel discovered the dangers of primacy in 1973 when it almost lost the Yom Kippur War to Egyptian and Syrian forces that it had handily defeated just six years before. The Israelis were caught off guard by new antitank and anti-aircraft missiles supplied by the Soviet Union—a foretaste of what the Information Age had in store.

In hindsight, the ability of the Egyptians and Syrians to bounce back from their humiliation in the Six-Day War (1967) should not have been so surprising. Defeat has often been a spur to innovation, from the Prussians' humiliation in the Napoleonic Wars, to the Germans' humiliation in World War I, to the Americans' humiliation in the Vietnam War. In the case of Japan in 1853, it did not take actual defeat but the mere threat of it, made explicit by the arrival of Commodore Perry's "black ships," to catalyze wide-ranging reforms. Out of all these setbacks were born new ways of fighting that led once-vanquished forces to victory on future battlefields.

It is much less common to see dominant powers innovating. More typical is the case of the Ottoman Empire, which mastered only one major military revolution—gunpowder—and then only in its early years. In their heyday in the fifteenth and sixteenth centuries, the Turks' gun-wielding armies and fleets carved out and defended a vast empire encompassing Asia Minor, North Africa, and the Balkans. By the eighteenth century, however, their glorious record of martial triumphs had become a major obstacle to making the innovations necessary to keep up with European competitors. The Sublime Porte's modernization was so belated and half-hearted that by the nineteenth century the onetime scourge of Christendom had become known as the "Sick Man of Europe." Early success set up the Turks, like so many others, for later defeat.

One of the few partial exceptions to this rule is Britain, whose Royal Navy stayed No. 1 from the age of sail to the age of steel. But not even the Royal Navy could successfully navigate the next major shift, from battleships to aircraft carriers—a failure that hastened the fall of the British Empire.

Business history is replete with the same story. Not a single maker of sailing ships made a successful transition to steam power. And not a single maker of minicomputers—not Digital Equipment Corporation, not Data General, not Prime, not Wang, all seen as invincible giants as recently as the 1980s—made a successful transition to personal computers. More recently, Sony, which was supremely successful in the era of transistor radios and portable cassette players, has struggled to adjust to the digital age, its Walkman having been superseded by Apple's iPod.

After a company fails, its employees, customers, and investors can always move on to a different firm. It is much harder for citizens to move on to another country after their homeland fails to keep up in the struggle for security. The stakes are obviously much greater in the geostrategic realm, making it all the more important to understand why some states fail and others succeed.

UNCONTROLLABLE CREATIVITY

History, alas, does not offer a blueprint of how the process of military innovation occurs. There is no single model that covers all cases, and this book has made no attempt to develop one. As James Q. Wilson noted in his magisterial study of bureaucracies:

Not only do innovations differ so greatly in character that trying to find one theory to explain them all is like trying to find one medical theory to explain all diseases, but innovations are so heavily dependent on executive interests and beliefs as to make the chance appearance of a change-oriented personality enormously important in explaining change. It is not easy to build a useful social science theory out of "chance appearances."

To the limited extent that we can generalize about five hundred years of history, it seems fair to say that the most radical innovations come from outside of formal military structures. There are some recent exceptions, such as the atomic bomb, the satellite, and the stealth airplane, but most of the key inventions that changed the face of battle since the Middle Ages—the cannon, handgun, three-masted sailing ship, steam engine, machine gun, rifled breech-loader, telegraph, internal combustion engine, automobile, airplane, radio, microchip, laser, wireless telephone—were the products of individual inventors operating more or less on their own: geniuses such as Robert Fulton, Hiram Maxim, Johann Nikolaus von Dreyse, and Guglielmo Marconi. Some had military applications in mind; most did not. For instance, the casting techniques that made cannons more effective in the fifteenth century were originally developed to make church bells.

Even where government has played a big role in the development process, as with the Internet and the electronic computer, the key advances were usually made by people not on its payroll: William Shockley, John Bardeen, and Walter Brattain (the transistor); Jack Kilby and Robert Noyce (the microchip); Ted Hoff (the microprocessor); Paul Allen and Bill Gates (MS-DOS and Windows); Tim Berners-Lee (the World Wide Web); Marc Andreessen and Eric Bina (the Mosaic browser); and many others. The laser is another case in point: It was invented in 1960 by Theodore Maiman, an unknown young physicist at Hughes Aircraft Company who had no Pentagon funding and who worked on the project despite, not because of, the wishes of his superiors.

While government and corporate R&D programs have grown exponentially since World War II, fundamental technological innovation (as opposed

to small-scale, incremental improvement) is simply too erratic and mysterious a process to be at the beck and call of any institution. "We can no more 'explain' the breakthroughs inside the minds of a Montgolfier or a Westinghouse," notes economist Joel Mokyr, "than we can explain what went on inside the head of a Beethoven when he wrote the *Eroica*."

Because creativity is so unpredictable, no country can count on making all, or even most, major scientific and technological breakthroughs.

DISSEMINATION AND NULLIFICATION

Moreover, few if any technologies, much less scientific concepts, will remain the property of one country for long. France matched the Prussian needle gun less than four years after the battle of Königgrätz. Germany matched the British *Dreadnought* three years after its unveiling. The USSR matched the U.S. atomic bomb four years after Hiroshima and Nagasaki. It is a truism that new technology, if it proves effective, tends to disseminate quickly.

One exception, and it is a notable one, is that technology was slow to move from the West to the rest of the world in the late stages of the Gunpowder Age and during the First Industrial Age. This created a yawning imbalance of power that allowed Europeans to conquer much of the world on the cheap. But by the mid-twentieth century the balance had righted itself, and Asians and Africans in possession of modern weaponry were able to win their independence from European states weakened by two world wars and the collapse of assumptions of racial superiority. Some analysts may discount the importance of technology in determining the outcomes of battles, but there is no denying the central importance of advanced weaponry in the rise of the West.

Military revolutions usually favor the offense at first, but, with the major exception noted above, that initial edge soon dissipates. The spread of *trace italienne* fortresses in the sixteenth century negated the advantage that heavy artillery had conferred on besieging forces. The spread of trenches starting in the American Civil War offset the impact of rifles and then machine guns. Today, sophisticated radar systems may be negating the edge enjoyed by first-generation stealth aircraft. More significantly, enemies are increasingly resorting to guerrilla tactics to avoid the full fury of American firepower.

The process of technological dissemination and nullification has speeded up since the rise in the mid-nineteenth century of such major arms manufacturers as Krupp, Winchester, and Armstrong, which were happy to sell to just about anyone. Thus, German troops were killed during the

Boxer Rebellion in 1900 by Chinese soldiers firing Mauser rifles and Krupp artillery pieces.

Contemporary arms manufacturers such as Lockheed Martin, Northrop Grumman, and the European Aeronautic Defense & Space Company operate under greater export restrictions but still seek to market the latest technology around the world. Even more pervasive today are firms that sell dual-use devices such as computers, night-vision goggles, and GPS trackers which can have both military and civil applications. Thanks to their success, many of America's key Information Age advantages are rapidly passing into the hands of friends and foes alike. This is part of a longer-term trend—the Westernization of the world increasingly puts the peoples of Asia, the Americas, and Africa on a par, economically as well as militarily, with those of Europe and North America.

As important as technological nullification is psychological nullification. The first time an army faces a major new weapon—the needle gun at Königgrätz, the machine gun at Omdurman, the tank in Poland and France, the smart bomb in the Gulf War—it is likely to be caught off guard. The resulting panic can be as damaging as the physical effects of the weapon itself. The next time, if there is a next time, the other side is likely to be less impressed. Thus the coalition bombing campaign of Iraq in 2003 did not induce the same "shock and awe" as its predecessor in 1991. Having been bombed more or less continuously for a decade, Iraqis had become inured to the effects of precision munitions. The speed and ferocity of the U.S. armored advance, by contrast, came as a surprise.

The way to gain a military advantage, therefore, is not necessarily to be the first to produce a new tool or weapon. Often it is to figure out better than anyone else how to utilize a widely available tool or weapon.

WEAPONS OF MASS DISRUPTION

Faster aircraft, heavier tanks, more accurate rifles, and longer-range artillery are all examples of what management guru Clayton Christensen calls "sustaining" innovations—i.e., products that don't require major organizational adjustments in order to assimilate. They are part of a process of continual improvement undertaken by most modern armed forces. But other innovations, the ones that are the focus of this book, are "disruptive" breakthroughs that occur less frequently and profoundly unsettle the status quo.

Taking advantage of these major advances usually requires what James Q. Wilson calls a "change-oriented personality"—someone like John Hawkins, Gustavus Adolphus, or Curtis LeMay who is not afraid to shake up

conventional ways of doing things. Fundamental changes can be preached from the outside but seldom imposed by civilians on a professional military. Consider the lack of success that J. F. C. Fuller and Basil Liddell Hart had in the 1930s preaching the gospel of armored warfare to the British army. John F. Kennedy had equal lack of success when he tried to get the U.S. Army interested in guerrilla warfare in the early 1960s—and he was the president!

The most successful innovators have tended to be people like Field Marshal Helmuth von Moltke, Admiral William Moffett, General Hap Arnold, and Generals Hans von Seeckt and Heinz Guderian: insiders, not outsiders. At best, civilians can play a supporting role in aiding military mavericks against their bureaucratic foes, though popular accounts tend to overstate the influence of flamboyant rebels such as Billy Mitchell. As political scientist Eliot Cohen writes: "To the extent that civilians can control the Department of Defense, they do so less by issuing edicts (which can be evaded, watered down, or delayed by the military and civil service bureaucracy) than by grooming and selecting internal leadership."

Change usually begins in peacetime, but it often takes a major battle—a Spanish Armada, Breitenfeld, Tsushima, Pearl Harbor, or Gulf War—to cause militaries to decisively shift how they do business. Sometimes battles take place during a time of great ferment but neither side can effectively implement a revolutionary operational concept, or both sides come up with the same concept, thereby forfeiting a chance to seize a decisive advantage. This was true of the U.S. Civil War and World War I, which is why both conflicts were such fearful, protracted slaughters and why neither one is considered at length in this book.

STRATEGY AND INNOVATION

Culture, geography, politics, and other factors greatly affect how receptive a military is to proposed changes. Especially important is a country's strategic situation, a combination of its location and its fears and ambitions. Geography is not destiny, or else it would be impossible to explain why Britain was a naval power for centuries while Japan—another island nation off the coast of a major continent—was not. Or why Prussia, rather than another nearby state such as Saxony or Bavaria, became a Great Power starting in the eighteenth century. Or why Sweden rose from obscurity to prominence in the seventeenth century and then fell back into obscurity in the eighteenth century—all without changing its geographic position. But, even though it is only one factor among many, there is no doubt that geography has influenced which nations are more receptive to which military revolutions.

Early in the modern period, Italian states like Venice were content to rely on galley fleets that served them well in the placid waters of the Mediterranean, the chief source of their foreign trade, but that proved utterly unsuited for a new way of naval warfare pioneered by Atlantic powers like Holland, Portugal, and England. Spain and France were caught in the middle, with one foot in the Mediterranean and another in the Atlantic. They never managed to become as effective at sail-and-shot tactics as the more single-minded English and Dutch. Centuries later, Germany became a leader in utilizing panzers because it planned to fight a fast-moving land war against numerous enemies on its frontiers. The Nazis did little to develop aircraft carriers or four-engine bombers because they did not think they needed them against their continental rivals. The United States was the mirror image: It led the way in the development of long-range bombers and aircraft carriers because it expected to fight a naval and air war against enemies far removed from its borders, but it did little to develop tank units because it did not expect to fight a major land war. Such expectations may turn out to be ill-founded (Germany could have used B-17s; the U.S. could have used Tiger tanks), but they powerfully affect decisions about the allocation of scarce resources.

It helps to have relatively few scenarios to prepare for. Germany in the interwar years had the luxury of preparing only for a land war in Europe, whereas Britain had to prepare not only for that contingency but also for naval wars in the Atlantic and Pacific as well as for imperial policing in its colonies. The U.S. had the advantage of focusing on a single foe after the Vietnam War. The concepts and technologies created to fight the Red Army just happened to be perfectly suited to battling the Iraqi army.

Today the U.S. faces a much bigger challenge because it has many potential foes, ranging from nonstate actors (al Qaeda and its ilk) to medium-sized powers (North Korea, Syria, Iran) and a rising great power (China). Because the U.S. has chosen to be strong in every sphere of combat (sea, land, air, space, cyberspace), in every type of warfare (from peacekeeping to high-intensity conflict), and in every corner of the globe, it faces pressure to invest and innovate in many different fields at once, or else to rein in its ambitions.

IS THERE A DEMOCRATIC ADVANTAGE?

Western states have been the most successful military innovators over the past five hundred years. There was something about western Europe (and its overseas offspring) that made it much more dynamic and open to change than other civilizations. Having a relatively liberal political and intellectual climate, of the kind that the West developed toward the end of the Middle

Ages, helps to create an atmosphere in which innovation can flourish. The Soviet Union's lack of freedom ultimately sabotaged its attempts to keep pace in the Information Age, just as the lack of freedom in Spain and France made it difficult for them to keep pace in a naval arms race with first the Netherlands and then Britain.

But we should be wary of simple-minded democratic triumphalism. History has offered many examples of autocratic states that proved more adept than their democratic rivals at exploiting military revolutions. The success of the Prussian and German armed forces between 1864 and 1942 and of the Japanese between 1895 and 1942 shows how well even relatively undemocratic systems can innovate. All that is required is some degree of openness to change, a commitment to meritocracy, and an ability to critically examine one's own mistakes—all disciplines in which the German General Staff, however illiberal, excelled. In fact, most democracies, which tend to be less militaristic than autocracies, face a disadvantage in taking advantage of military innovations because they are less inclined to be generous to their armed forces in peacetime—a problem that beset liberal Western nations during the 1930s.

Nor is there much evidence to suggest that soldiers fight better for a democracy than for a dictatorship. Man for man, the Wehrmacht was probably the most formidable fighting force in the world until at least 1943, if not later. German soldiers were even known for showing more initiative than the soldiers of democratic France, Britain, and the United States. Meanwhile, Soviet soldiers stoically endured privations and casualties far beyond anything suffered by their Western allies. North Vietnam is another modern state that fielded superb armies despite a notable democracy deficit.

In any case, the differences between the armies of dictatorships and democracies are less significant than they may appear at first blush. Even the most liberal states have to employ command-and-control methods in their armed forces. And even the most autocratic states have to pay attention to troop morale and to allow room for individual initiative.

But if democracies do not have an advantage in creating formidable war machines, they do seem to have an intrinsic edge in figuring out how to use them. Autocracies tend to run amok because of the lack of internal checks and balances. Philip II, Gustavus Adolphus, Louis XIV, Frederick the Great, Napoleon, Wilhelm II, Hitler, the Japanese leaders of the early twentieth century—they all built superb militaries but ultimately led their nations into ruinous wars. (So did Saddam Hussein with his less impressive but nevertheless formidable army.) They had no sense of limits, and no politician was strong enough to stop them. Their tactics may have been superb, but their grand strategy was lousy, the best examples being Napoleon's and Hitler's foolhardy invasions of Russia. Democracies sometimes overreach too (witness the Boer, Algerian, and Vietnam wars), but they tend to avoid the worst traps be-

cause they have a more consensual style of decision-making. As two political scientists note, "being vulnerable to the will of the people restrains democratic leaders and helps prevent them from initiating foolhardy or risky wars."

BUILDING BETTER BUREAUCRACIES

The key to successful innovation, whether for a dictatorship or a democracy, is having an effective bureaucracy. This was the chief advantage enjoyed by Elizabeth I over Philip II, George III over Daulat Rao Sindia and Raghuji Bhonsle II, Emperor Meiji over Czar Nicholas II, Adolf Hitler over Édouard Daladier and Paul Reynaud, and the two George Bushes over Saddam Hussein. Prussia's secret weapon in the nineteenth century was not the needle gun or the railroad or the telegraph. It was the General Staff, which figured out how to utilize these innovations.

America's secret weapon today is not the stealth airplane or the Predator but the agency that was responsible for their development (and much else besides)—the Defense Advanced Research Projects Agency. Ever since its forerunner was set up in 1958 during the Sputnik crisis, DARPA has shown how a government agency can push the frontiers of innovation by allocating grants to universities, think tanks, and private companies for high-risk ventures. To the limited extent that innovation can be systematized, DARPA has done it. Other nations trying to compete with the United States are hobbled by not spending as much as the U.S. does on research and development, but even spending more money would be no guarantee of success. If it were, then the European Union, whose collective defense budget is two-thirds the size of America's and which has even more soldiers under arms, would be much closer to the U.S. in military capabilities than it actually is. The problem is that most European spending is unfocused, duplicative, and inefficient, whereas DARPA has been smart about allocating its \$2 billion annual budget.

Bureaucracies are so important because, as this book has stressed, the realization of a Revolution in Military Affairs requires far more than simply revolutionary technology. It also requires revolutions in organization, doctrine, training, and personnel. That was achieved by the Swedes when they crafted mixed-arms formations made up of pikemen and musketeers; by the Prussians when they figured out how to rapidly mobilize and move large numbers of riflemen by railroad; by the Japanese when they decided to group aircraft carriers together in strike groups; and by the Americans when they integrated smart bombs, sensors, stealth, and professional soldiers in the AirLand Battle doctrine.

Bureaucratic innovation can seldom be limited to the military alone

because armed forces are always a reflection, however refracted, of the broader society. Each military epoch comes with its own distinctive system of governance. The rise of the Gunpowder Age fostered the growth of absolute monarchies. The First and Second Industrial Ages fostered giant welfare and warfare states. The Information Age is leading to a more decentralized, flatter form of government and the rise of more powerful nongovernmental groups. States that fail to keep up with these transformations risk getting run over by those that do.

For powers that lack effective bureaucratic structures, the possession of modern weaponry is of dubious utility, as states from eighteenth-century India to the twentieth-century Middle East have found. The Arab nations are particularly egregious in this regard: Their long record of military futility since 1945 comes despite having access to copious stocks of modern arms from such outside suppliers as France and the Soviet Union. No matter how great the Arab preponderance in men and materiel—and against Israel in 1948, 1956, 1967, and 1973 their advantage appeared, on paper at least, to be insuperable—they have continually contrived to snatch defeat from the jaws of victory. In one of the lesser-known episodes of this long record of ignominy, the well-armed Libyan military was routed by ill-armed Chadians in 1986–87 after Moammar Qadafi tried to annex northern Chad. The only military strategies (if such they are) that Arabs have been able to employ with any success are terrorism and repression. “The incompetence of Arab tactical leadership, their severe problems managing information, and the inability of their personnel to properly employ and maintain their military hardware,” writes former CIA analyst Kenneth M. Pollack, “left the Arab states highly vulnerable to most potential adversaries.”

The armies of Russia and the United States were far more competent, but in Afghanistan and Chechnya, Vietnam and Iraq, they, too, found themselves stymied by smaller, poorer adversaries, largely because their armed forces were not properly configured for counterinsurgency warfare. This does not mean that modern military hardware is useless—only that by itself it is not enough to guarantee victory against a clever, determined adversary. When combined with the right organization, doctrine, training, and leadership, however, sophisticated weaponry can confer a decided advantage even in battling irregular foes.

DREAD OF INNOVATION

It is no surprise that the authoritarian Arab states have not, for the most part, managed to make the changes necessary to harness modern military

power. No Arab dictator can afford to have a military that is too strong for fear that it will be employed against him. But even for more liberal polities, which generally need not fear a military coup d'état (though France faced such a prospect as recently as the early 1960s), transitions from one military system to another can be wrenching, because they require uprooting existing career patterns and deeply held beliefs. Officers trained in cavalry charges were not happy about the advent of tanks, any more than sailors trained in battleships were happy about the arrival of aircraft carriers, or knights trained in sword fighting were happy about the spread of firearms. Militaries are inherently conservative organizations. As a British army colonel noted in 1839: “In no profession is the dread of innovation so great as in the army.”

Successful adaptation to major technological shifts requires overcoming that dread and changing the kinds of people who are rewarded within a military structure. The rise of railroads and steamships in the nineteenth century elevated the importance of logisticians and engineers, technocrats who were initially despised by traditional army and navy officers. In the Imperial German Navy, engineering officers were sent to a separate school, they wore less gaudy uniforms, without the sashes and imperial crowns sported by regular officers, and they were not allowed to dine in the officers' mess. They even had to endure the indignity of having their wives addressed as “women,” not “ladies.” It was not until 1899, half a century after the adoption of steam power, that the U.S. Navy merged “line” and “engineering” officers. In the more aristocratic British and German navies the process took even longer.

Today, the U.S. Air Force faces difficult dilemmas as it figures out how to integrate unmanned aerial vehicles: Should someone controlling a Predator from a trailer thousands of miles away be a certified pilot? Should control time count as “flying” hours? This may seem a picayune matter, but it looms large in a service where the fastest advancement has always gone to fliers. If the air force doesn't give greater promotion opportunities to UAV operators, it cannot attract and keep the best people for these jobs, but if it advances them its fighter-jock culture will inevitably change.

This is part of a broader challenge confronting all Information Age militaries: how to make room for those who fight with a computer mouse, not an M-16. Will traditional warriors—men with shaved heads and hard bodies—continue to run things, or will nerds with bad posture and long hair, possibly even women, assume greater prominence? Two Chinese strategists write that “it is likely that a pasty-faced scholar wearing thick eyeglasses is better suited to be a modern soldier than is a strong young lowbrow with bulging biceps,” but, even if that is true, reordering any military along those lines presents a far more profound and problematic challenge than questions about which tank or helicopter to buy. As Eliot Cohen writes: “The cultural challenge for military organizations will be to maintain a warrior

spirit and the intuitive understanding of war that goes with it, even when their leaders are not, in large part, warriors themselves."

THE DANGER OF TOO MUCH CHANGE—AND TOO LITTLE

This book surveys many instances where militaries had to change or die. Those armed forces that did not successfully integrate the gun, the long-range bomber, precision-guided munitions, or other important innovations experienced the agony of their members dying in great numbers. But we have also looked at some instances of militaries too eager to change in the wrong way. In the 1930s, the U.S. Army Air Corps and the Royal Air Force placed too much faith in the ability of unescorted bombers to win a future war—a doctrinal mistake that cost the lives of tens of thousands of air crews over Europe. In the 1940s Hitler poured vast resources into the development of the V-1 and V-2, resources that might better have been employed on his conventional forces. And in the 1950s the U.S. Army, Navy, and Air Force did so much to rearrange themselves around the demands of the nuclear battlefield that they were not ready for the actual threat they would soon confront in the jungles of Vietnam.

Arguably a similar phenomenon has occurred in Iraq, where the Information Age armed forces of the United States have found themselves frustrated by less sophisticated adversaries. Many now ask: Why did the Defense Department not invest in more linguists, more MPs, more civil affairs specialists, more soldiers in general, rather than in more JDAMs and JSTARS? The answer is that senior leaders, such as Donald Rumsfeld, believed that the future of warfare lay in high-tech information systems, not in lowly infantrymen. This does appear to be a mistake in light of events in Iraq—but it may not turn out to be so mistaken if the U.S. finds itself in a clash with China or North Korea.

There is no rule of thumb to suggest how much or how little a military should change in response to technological developments. Each revolution raises painful questions of prioritization such as those that the United States and other countries confront today: Should they pay for more traditional infantrymen, or push resources into "transformational" programs like surveillance satellites, wireless broadband networks, and directed-energy weapons? Should they continue to build traditional tanks, aircraft, and ships or switch to unmanned platforms? Each path has major risks and trade-offs: Paying for larger standing forces can make it easier to respond to today's threats; cutting force strength and using the savings to pay for high-tech hardware

can make it easier to respond to tomorrow's threats. (The only strategy that definitely won't work is the one that many European countries are pursuing: cutting existing forces but not investing the savings in any other defense program.) It would be nice to be able to do everything at once. But no one, not even the Pentagon, has enough money for that.

History indicates that the wisest course is to feel one's way along with careful study, radical experimentation, and freewheeling war games. Paradoxically, revolutionary transformation often can be achieved in evolutionary increments. The Germans did not shift over their entire army to panzer divisions in the interwar years. In 1939–40 only about 10 percent of German forces were composed of armored units, and the Wehrmacht had more ponies than panzers, but this was enough to produce breakthroughs from Poland to France. Likewise, General H. H. Kitchener did not have many machine guns when he confronted the Mahdists at Omdurman, but the few he had produced utterly devastating results. Nor did the U.S. in the 1980s convert its entire air force to stealth aircraft, but having even a few F-117s had an outsize impact on the outcome of the Gulf War. A little cutting-edge technology can go a long way against a less advanced foe.

A corollary is that a military revolution does not necessarily sweep aside all old weapons and old ways of doing things. Battleships may have been dethroned as queens of the sea in 1941, but they continued to perform a valuable auxiliary role as a shore bombardment platform into the 1980s. Gustavus Adolphus did not simply toss out pikes in order to make way for muskets and cannons; he used a combination of weapons, old and new, to achieve the maximum shock effect. Indeed, bayonets continue to be fixed onto rifles (though rarely used) hundreds of years after edged weapons lost their primacy on the battlefield. And nuclear weapons have hardly rendered conventional weapons obsolete.

This offers a counterpoint to skeptics who deny the existence of an Information Revolution simply because not everything has changed: It never does. On the other hand, this also offers a cautionary lesson that some modern-day J. F. C. Fullers or Billy Mitchells anxious to scrap the tank, the aircraft carrier, or the manned airplane should keep in mind: Introducing "transformational" systems does not necessarily mean getting rid of all "legacy" platforms. Rather, it means readjusting the balance between the two. "You need to think about how to make a transition," counsels Andrew Marshall of the Pentagon's Office of Net Assessment, "not about how to eliminate current weapons."

While no one would wish for more combat, the U.S. armed forces are helped in this process by having so many wars to fight that can serve as field laboratories for the testing of new technologies. The first Predator was rushed into service for the Kosovo War. Having performed well there, an armed

version made its debut two years later in Afghanistan. This, in turn, spurred the development of purpose-built unmanned combat aerial vehicles that will no doubt be tried out in another conflict before long.

The U.S. armed forces would be even more adept at innovation if they were willing to stage more realistic war games in which adversaries could use unconventional tactics instead of fighting in the way that American admirals and generals would prefer. They would also be helped if defense spending could be allocated according to a rational judgment of strategic priorities, not based on the political muscle of major defense contractors and their allies on Capitol Hill. That, however, seems unlikely to change as long as America remains a democracy.

SILENT SPUTNIK?

The quickening pace of scientific and technological progress, which has been speeding up ever since the start of the Industrial Revolution more than two hundred years ago, puts a greater premium on having scientists and engineers who can stay at the frontiers of change. It is neither necessary nor possible to invent everything yourself. But it is vital, at a minimum, to be able to keep up with breakthroughs made elsewhere and take advantage of their military potential. The Royal Navy did not have to be the technological leader in the nineteenth century, but it did need the capability to quickly match and surpass any innovation by the No. 2 naval power, France.

In this regard, many experts note that U.S. hegemony might be endangered by its failure to produce more math, science, and engineering graduates. The U.S. has made up for this shortfall in the past by importing smart people from abroad (38 percent of science and engineering doctorate-holders in the U.S. are foreign-born). But that has become harder to do in the wake of post-9/11 visa restrictions and booming economies in eastern Europe, India, China, South Korea, and Taiwan—the major sources of American scientific and engineering talent—that discourage immigration to America. If China can keep more of its geniuses at home, it will be easier for Beijing to challenge U.S. power. Some scientists warn that the U.S. is facing a “Silent Sputnik” crisis which could imperil U.S. leadership.

Remedying this looming shortfall will probably require more funding for math, science, and engineering education, and that won’t be cheap. It will be even more expensive to translate the resulting ideas into actual military programs. It does not necessarily take a lot of money to innovate; breakthroughs like the blitzkrieg and carrier warfare emerged out of paltry military budgets in the interwar years. But it does take a lot of money to

bring inventions to fruition, especially today, when each new weapons system costs several times more than its predecessor. It also costs a lot to field high-quality soldiers able to cope with the complexity of modern war. The annual cost to the United States for each member of its armed forces more than doubled in constant terms over the past thirty years—from \$125,000 per person in 1970 to \$264,000 per person in 2003—and it continues to increase.

There is no free lunch: Despite the fervent hopes of some transformation advocates, there is no way to significantly increase military power while cutting costs. Today, even more than in the fifteenth century, more military capability requires “money, more money, and again more money.” With the U.S. facing budget deficits and looming bills for social welfare programs, questions inevitably arise about whether it can afford to keep spending so much on defense. Other countries confronting tight budgets face similar debates: Is it necessary to stay at the forefront of military change? Is it vital to take advantage of the Information Revolution and its successors?

The answer will vary from country to country. As one security analyst notes, “try as they might, countries like Burkina Faso or Paraguay . . . will never be candidates for the exploitation of the RMA.” But for the United States and other countries that aspire to be first- or second-tier military powers, there is no alternative but to stay abreast of the changes—tactical and technological, conventional and “asymmetric”—transforming the modern battlefield.

WHY RMA’S MATTER

The major theme that runs throughout this book is the importance of not missing out on the next big change in warfare. History is driven by many factors, but while academia focuses on economics, race, class, sexuality, geography, germs, culture, and other influences on the course of human events, it would be foolish and short-sighted to overlook the impact of military prowess and especially aptitude in taking advantage of major shifts in war-fighting. Of course, a country’s success or failure in harnessing change cannot be divorced from such underlying factors as its economic health, scientific sophistication, educational system, political stability, and so forth. But, contrary to Napoleon, God is not necessarily “on the side of the big battalions.” Even big and wealthy countries often lose wars and head into long-term decline through a lack of military skill.

The considerable gains of the Axis during the early years of World War II came, after all, against a coalition of Allied states that in aggregate had 40

percent greater GNP and 170 percent larger population. That the Axis ultimately lost shows that military skill can sometimes be trumped by greater resources if a war drags on long enough *and* if the side with greater resources shows sufficient wisdom in their employment. But even in a long coalition war, the side with the greater resources does not always prevail. The alliance of Britain, Hanover, and Prussia was dwarfed in economic and demographic resources by its adversaries in the Seven Years' War (1756–63)—France, Austria, Russia, Sweden, Saxony, and (near the end) Spain—but still managed to win largely because of the superior skill of the Prussian army and the British navy. This book chronicles many other examples of the poorer side emerging victorious: Britain beat the Spanish Armada, Sweden beat the Holy Roman Empire at Breitenfeld and Lützen, Prussia beat the Habsburg Empire at Königgrätz, and Japan beat the Russian Empire in 1904–1905. More recent instances might be cited, such as North Vietnam's defeat of the United States or the Afghans' defeat of the Soviet Union.

These were not anomalies. In a statistical analysis of twentieth-century wars, the side with the larger GNP, population, armed forces, and defense expenditures won only a little more than half the time, making these factors about as useful in predicting military outcomes as flipping a coin. Political scientist Stephen Biddle, who analyzed these statistics, writes, "Superior numbers can be decisive or almost irrelevant depending on the two sides' force employment. This in turn means that states' relative economic, demographic, or industrial strength are poor indicators of real military power: gross resource advantages matter only if they can be exploited via modern-system force employment, and many states cannot do so. . . . How forces are used is critical."

Indeed, while some states translate riches into military power, as the U.S. did in the early years of the twentieth century, just as many states translate military power into riches. That is what England did when it sent its navy to conquer colonies and carve out trade routes, and what Prussia did when it sent its army to overrun the neighboring German principalities. Some states are drained by war, but many others attain Gustavus Adolphus's ideal of making war "pay for itself"—a feat achieved most recently by the United States when it succeeded in making its allies foot most of the bill for liberating Kuwait in 1991.

The ongoing proliferation of destructive technology means that the link between economic and military power is more tenuous than ever. Al Qaeda, whose entire budget would be insufficient to buy a single F-22, can inflict devastating damage on the world's richest country. Advances in biological and cyberwar promise to put even more destructive potential into the hands of ever smaller groups, as does the continuing proliferation of nuclear weapons.

Imagine the devastating consequences of a mega-attack by terrorists. Not only could millions die, but international travel and commerce—the lifeblood of the global economy—could be severely disrupted. Such a scenario reveals the falsity of economic determinist arguments which counsel that military strength is unimportant and that it is feasible to stint on military preparedness in order to strengthen the economy. On the contrary, there can be no long-term prosperity without security. The entire world today depends, no matter how begrudgingly or unwittingly, on the protection provided by the United States, whose armed forces keep open air and sea lanes, safeguard energy supplies, and deter most cross-border aggression.

Dreamers can convince themselves that military power no longer matters, that economic interdependence has consigned war to the dustbin of history, that a country need wield only "soft power," but history is likely to deliver a stark rebuke to such wishful thinking. As a matter of fact, it already has. The attacks of September 11, 2001, put an end to a decade of talk about the "end of history," a "strategic pause," the inexorable flow of "globalization," and the "peace dividend." The incidence of war may have declined for the moment, but great dangers still loom ahead. Santayana had it right: "Only the dead have seen the end of war."

FIGHTING WILDCATS AND RODENTS

Technological advances will not change the essential nature of war. Fighting will never be an antiseptic engineering exercise. It will always be a bloody business subject to chance and uncertainty in which the will of one nation (or subnational group) will be pitted against another, and the winner will be the one that can inflict more punishment and absorb more punishment than the other side. But the way punishment gets inflicted has been changing for centuries, and it will continue to change in strange and unpredictable ways.

In assessing the future conduct of conflict, most analysts tend to fall into one of two camps. One group stresses the dangers of terrorists and guerrillas who use cheap, simple weapons like AK-47s, machetes, or explosives. Another group stresses the danger of high-tech weapons such as cruise missiles and killer satellites proliferating around the world and into the hands of states such as China and North Korea. The former school (associated with ground-combat arms) stresses the need for better warriors; the latter school (associated with air and naval forces), the need for better machines. The reality is that both high-intensity and low-intensity threats are real and that both top-notch people and first-rate equipment are needed to counter them.

Michael Evans of the Royal Military College of Australia offers sage advice when he writes: "In a dangerous and unpredictable world, military professionals and their political masters must . . . be ready to tame the big wildcats and not simply the vicious rodents, to be able to fight troops like Iraq's former Republican Guard as well as Taliban, al-Qa'ida militia, and terrorists."

Today, the U.S. is much farther along in figuring out how to tame the Republican Guard than al Qaeda, and it needs to place more emphasis on making up for its deficiencies in irregular warfare rather than simply enhancing its already substantial lead in conventional warfare. While the Information Revolution has decreased the number of weapons and soldiers needed to defeat a conventional adversary, occupation duty and nation-building—the prerequisites for turning a battlefield triumph into a long-term political victory—continue to demand lots of old-fashioned infantry. Therefore, the U.S. and its allies would be making a mistake if they were to seriously stint on force size in order to procure more high-tech systems.

But that doesn't mean that the U.S. can ignore the dangers of major warfighting or the dictates of technological change. That was the mistake Britain made before 1914 and again before 1939. The British had the world's best "small war" force—an army well-trained and equipped for fighting bandits and guerrillas—but it was ludicrously insufficient to deter German aggression or to defeat Germany once a world war broke out. That mistake, symbolized by deficiencies in tanks and aircraft carriers, hastened the end of the Pax Britannica.

Today, the possibility of conventional interstate war is lower than at any time in the past five hundred years, but it has not disappeared altogether. Because Americans and other citizens of Western democracies no longer seem willing to suffer the same level of casualties experienced by their ancestors, their armed forces must be able to defeat adversaries at scant cost in lives. That argues for keeping the qualitative edge that the U.S. gained in the Information Age—an edge that cannot be preserved by standing still. It will be necessary to keep innovating because, as previously indicated, some of the technologies and techniques employed by the U.S. are starting to be negated by their dissemination around the world. Innovation must be organizational as much as technological, and it needs to focus on potential threats across the entire spectrum, from low-intensity guerrilla wars to high-intensity conventional conflicts.

In any case, the boundaries between "conventional" and "unconventional," "regular" and "irregular" warfare are blurring. Even nonstate groups are increasingly gaining access to the kinds of weapons—from missiles and land mines to chemicals and perhaps even atomic bombs—that were once the exclusive preserve of states. And even states will increasingly turn to unconventional strategies to blunt the impact of American power.

Two colonels of the Chinese People's Liberation Army envision "unrestricted warfare" encompassing not only traditional force-on-force encounters but also financial warfare (subverting banking systems and stock markets), drug warfare (attacking the fabric of society by flooding it with illicit drugs), international law warfare (blocking enemy action by using multinational organizations), resource warfare (seizing control of vital natural resources), even ecological warfare (creating man-made earthquakes, tsunamis, or other disasters). In a clever bit of jujitsu, many of these strategies turn the strengths of Information Age countries against them. Al Qaeda is pursuing similar strategies.

Countering such threats will require much more than simply buying more advanced aircraft, tanks, or submarines. Such traditional weapons systems may be almost entirely useless against adversaries clever enough to avoid presenting obvious targets for precision-guided munitions. To fight and win the wars of the future—wars that may more closely resemble a series of terrorist attacks or hit-and-run raids than traditional force-on-force armored, aerial, or naval engagements—will require reorganizing conventional militaries to emphasize such skills as cultural awareness, knowledge of foreign languages, information operations, civil affairs, and human intelligence. It will also require cutting away the bureaucratic fat to turn bloated Industrial Age hierarchies into lean Information Age networks capable of utilizing the full potential of high-tech weapons and highly trained soldiers.

The United States' readiness for such challenges will determine whether it can keep its position as the lone superpower or whether the world will see another power shift of the kind that accompanied the Gunpowder Revolution, the First Industrial Revolution, the Second Industrial Revolution, and the early stages of the Information Revolution. The course of future history will turn on the outcome.